Daily Briefing - November 01, 2026

Your Daily Tech & Programming Digest

Sunday, November 01, 2026

1000	121729	1182	36
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** <p class="">We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Automated multiscale high-content imaging connects plaque population heterogeneity to localized synapse loss

👤 Polino, C., Port, R., Hofmann, M., Smith, S., Wang, X., Klein, R., Marino, M. 📅 2026-10-31 📖 1 min

📝 159 words BIORXIV NEUROSCIENCE

> **Summary:** Synapse loss is a major correlate of cognitive decline in neurodegenerative disease, yet its relationship to localized pathology remains difficult to study at scale. Conventional confocal imaging approaches are limited by low throughput, selection bias, and restricted sampling of pathological featur...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761378v1?rss=1

The Canadian Pediatric Imaging Platform (C-PIP) Reference Sample: Neuroimaging Protocol and Cross-site Validation for Harmonized Multimodal Pediatric Imaging

👤 Singh, M., Parimoo, S., Enguix, V., Cortese, F., Camacho, M., Spinney, S., Lynch, S. J., Carlson, H. L., Leijser, L. M., Tardif, C. L., Leppert, I. R., Pike, G. B., Kadalie, E., Near, J., Kumaragamage, C., Chitroda, R., Lauzon, M. L., Manning, K. Y., Varma, G., Paus, T., Harris, A. D., Lodygensky, G. A., Beauchamp, M. H., Mabott, D. J., Kopala-Sibley, D. C., Vorstman, J. A. . S., Yeates, K. O., Arnold, P. D., Jacquemont, S., Kadis, D. S., Lippe, S., Bellec, L., Lebel, C., Conrod, P., Bray, S., Wheeler, A. L.

📅 2026-10-31 📖 1 min 📝 222 words BIORXIV NEUROSCIENCE

Summary: Understanding how the human brain matures from infancy through adolescence is essential for identifying factors that influence health outcomes. Magnetic resonance imaging (MRI) provides powerful tools for characterizing structural, microstructural, functional, and metabolic aspects of brain developm...

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https://www.biorxiv.org/content/10.64898/2026.10.26.761379v1?rss=1

Recurrent Spiking Neural Networks for Continuous Motor Control: A Biologically Plausible Model of Motor Cortex

👤 Imamura, H., Imamura, F., Igarashi, J., Isomura, Y., Hira, R.

📅 2026-10-31 📖 1 min 📝 216 words

BIORXIV NEUROSCIENCE

Summary: Motor cortex acquires skilled movements through recurrent spiking dynamics and experience-dependent synaptic plasticity, yet how such circuits learn continuous motor control using biologically local signals remains unclear. Here, we develop a recurrent spiking neural network model of motor cortex th...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.756490v1?rss=1

Intelligent clinical decision support system for intracranial aneurysm lesion-based detection using biomedical imaging data

📅 2026-11-01 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76815-z

Cortex-wide neuronal dynamics underpin quasi-periodic patterns in resting-state fMRI

📅 2026-11-01 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11228-2

Network geometry shapes the transformation of task representations across the human cortex

👤 Michael W. Cole 📅 2026-10-22 📖 1 min 📝 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02486-8

Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1938329

Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

2026-10-14 1 min 0 words FRONTIERS NEUROROBOTICS

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.2023207

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Emerging light-driven neuromorphic hardware for artificial intelligence

👤 Dimitris Tsoukalas 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Systemic, RPE-directed AAV-tyrosinase therapy restores ocular pigmentation in an OCA1 mouse model

👤 Leah C Byrne 📅 2026-10-31 📖 1 min 📝 71 words LOW VISION

Summary: Oculocutaneous albinism type 1 (OCA1) is a pigmentation disorder caused by biallelic tyrosinase (TYR) mutations. TYR inactivity results in loss of hair, skin, and eye pigment, which is detrimental for ocular function. Hypopigmentation of iris, retinal pigment epithelium (RPE), and choroid results in...

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Systematic review of atropine cost-effectiveness in childhood myopia management

Long Chiau Ming 2026-10-31 1 min 45 words LOW VISION

Summary: CONCLUSION: Studies demonstrate favourable economic value using diverse metrics, including cost per diopter reduction, cost ratios, and cost-utility estimates, rather than uniformly reported ICERs. Greater economic benefits are observed over longer time horizons, largely driven by the avoidance of h...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

Indri Febriani 2026-10-31 1 min 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

Katharina Brugger 2026-10-31 1 min 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

Chuxu Wang 2026-10-31 1 min 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang 📅 2026-10-29 📖 1 min 📝 39 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu
2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

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Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

Shouyan Wang 2026-10-30 1 min 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

> **Summary:** CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

> **Summary:** CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

👤 Jianfeng Hu
📅 2026-10-31 📖 1 min 📝 59 words

BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

Jing Jin 2026-10-31 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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Monthly Updates [October]

2026-10-01 5 min 1097 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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ChatGPT and Codex usage limits merging

👤 Zariel 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

Summary: Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

> **Summary:** In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361153

Table of Contents

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

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IEEE Engineering in Medicine and Biology Society

📅 2026-01-29 📖 1 min 📝 1 words REVIEWS BIOMEDICAL ENGINEERING

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Front Cover

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/uncategorized/welcoming-the-newly-elected-embs-adcom-members-for-2026-2028/

Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming & Congratulating Our Newly Elected EMBS ExCom Members appeared first on IEEE EMBS.

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https://www.embs.org/uncategorized/welcoming-congratulating-our-newly-elected-embs-excom-members/

EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

> **Summary:** Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

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https://www.embs.org/featured-news/ieee-pulse-editor-in-chief-search-submit-and-application/

Metformin alleviates neuronal injury in MK-801-Induced Schizophrenia-Like mice by inhibiting neuronal ferroptosis

📖 1 min 📝 25 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yanwen Lin, Gaoxia Wen, Xiaoyue Hu, Ran Hu, Yahong Duan, Na Wu, Peng Xiong, Xianlai Jin

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https://www.sciencedirect.com/science/article/pii/S0006899326004610?dgcid=rss_sd_all

Distinctive patterns of right-hemisphere vulnerability in cerebral small vessel disease: a gray matter asymmetry study

📖 1 min 📝 15 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Jingdian Lu, Yanli Wu, Lei Gao

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N-methyl-D-aspartate receptors in the prelimbic cortex modulate chronic neuropathic pain and comorbid generalised anxiety in a rainbow boa snake threat paradigm

📖 1 min 📝 38 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Priscila Medeiros, Ana Carolina Medeiros, Ohana Turcato Macacare, Weverton Castro Coelho-Silva, Renata Moreira Acunha, Fernando René Bendaña-Córdoba, Farhad Ullah, Sérgio Henrique Ferreira, Renato L...

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Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

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Zona incerta circuits in cognition and the regulation of sleep and wakefulness: potential relevance to Alzheimer’s disease

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Jun Kang, Yang Li, Wenjun Yan

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State transitions in amyotrophic lateral sclerosis: mechanistic convergence and biomarker-guided stratification

📖 1 min 📝 23 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Mo Nasb, Dan Yang, Tayyab Hamid Malik, Dandan Xu, Lamis Dayoub, Jing Zhou, Yan Zhao

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https://www.sciencedirect.com/science/article/pii/S0306452226007062?dgcid=rss_sd_all

Dynamic multi-scale topology learning and cross-modal graph fusion for autism spectrum disorder diagnosis

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Huadong Zhang, Junyong Zhao, Shaoqi Wu, Krisztián Szigeti, Kellermayer Miklós, Jiande Sun, Jianping Qiao

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Diagnostic classification of ASD improves with positively sparsified DTI inter-regional correlation and graph-theoretical artificial intelligence

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ravi Ratnaik, Sriram Kumar P., Lavanya Selvaganesh, Sandeep Singh Sengar, Jac Fredo Agastinose Ronickom

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Decoding individual differences in mental information from human brain responses predicted by deep neural networks

📖 1 min 📝 20 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kiichi Kawahata, Jiaxin Wang, Antoine Blanc, Naoya Maeda, Shinji Nishimoto, Satoshi Nishida

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006270?dgcid=rss_sd_all

Beyond rumination: an ecological momentary assessment of daily thinking patterns in maladaptive daydreaming

👤 Lu, A. S., Reich-Fuehrer, M., Andrews, E., Hennessy, S., Christoff Hadjiilieva, K., Kam, J., Andrews-Hanna, J.

📅 2026-10-31 📖 1 min 📝 260 words BIORXIV NEUROSCIENCE

> **Summary:** Maladaptive daydreaming is a form of maladaptive repetitive thought characterized by excessive, dysfunctional fantasy engagement, with notable links to psychological distress and poor well-being. Although symptoms can be reliably measured along a continuum, it remains unclear how they manifest in th...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.29.762728v1?rss=1

Structural Evolution of Neural Networks during Zebrafish Early Development

Wang, Z., Tian, L., Li, B.
2026-10-31 1 min 144 words BIORXIV NEUROSCIENCE

Summary: The spatial architecture of neural networks underpins nervous system function, yet dynamic in vivo evidence for their early establishment process remains scarce. Here, using long-term confocal live-cell imaging of transgenic zebrafish, we dissected neural network development across the brain, yolk s...

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https://www.biorxiv.org/content/10.64898/2026.10.29.762732v1?rss=1

Circuit mechanisms for disentangled representations and generalization in the hippocampus

Antonio Fernandez-Ruiz
2026-10-27 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02491-x The hippocampus encodes different aspects of experience as disentangled variables, enabling flexible recombination and generalization to new s...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02491-x

Stimulus-specific recruitment of human amygdala neurons predicts episodic memory encoding success

👤 Ueli Rutishauser
📅 2026-10-27 📖 1 min 📝 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 27 October 2026; doi:10.1038/s41593-026-02493-9 The study shows that stimulus-specific neurons in the human amygdala determine which experiences become lasting memory, revealing a mechanism ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02493-9

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

👤 Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Wave-function formalism for evolving neural dynamics: a mathematical framework informed by EEG

👤 Madelyn Klebba 📅 2026-10-12 📖 1 min 📝 268 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The organization of distributed neural activity remains one of the central challenges in computational neuroscience. Contemporary frameworks, including Integrated Information Theory, Global Workspace Theory, entropy-based approaches, and electromagnetic field theories, have advanced our understandin...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1812909

Correction: Time-course profiling of behavioral phenotypes and Ywhah/ERK signaling expression within the vestibular region of the brainstem during motion sickness habituation

👤 Zhanguo Jin 📅 2026-10-29 📖 1 min 📝 0 words

FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2026774

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

👤 Kwadwo Owusu Akuffo
📅 2026-10-30 📖 1 min 📝 51 words LOW VISION

> **Summary:** CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

👤 Faisal Hamed Aljamea
📅 2026-10-30 📖 1 min 📝 53 words LOW VISION

> **Summary:** CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard 2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe 2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Adaptive multiscale optical spatial filtering via a polarization-tunable phase-change micropore array

Yikai Chen 2026-10-30 1 min 62 words LOW VISION

Summary: As artificial intelligence and machine vision tasks evolve toward high throughput and low latency, all-optical analog signal processing demonstrates tremendous potential. However, the optical responses of most conventional static structures are fixed, rendering dynamic reconfiguration challenging. D...

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Visual acuity measurements in monochromatic and low-blue illumination

👤 Dragan Sekulovski 📅 2026-10-30 📖 1 min 📝 76 words LOW VISION

Summary: It is well known that visual acuity increases with light intensity. We report on the possibility to further increase visual acuity by changing the spectrum of the light while keeping light intensity fixed, or alternatively, provide the same visual acuity at lower light intensity. For a broad-spectru...

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Sensor-aware physical modelling for deep learning-based single-photon imaging

👤 Yu Pan 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Deep learning-based single-photon imaging commonly relies on synthetic time-resolved histograms with incompletely specified sensing physics. We present a sensor-aware framework that represents the source-receiver response, spatially varying background, detector artifacts, and first-detection acquisi...

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FDUNet-LRSA: efficient frequency-adaptive segmentation for automated crop disease inspection

Kai Huang 2026-10-31 1 min 56 words LOW VISION

Summary: The transition toward industrialized and automated agriculture requires reliable machine vision for high-throughput crop health monitoring and field decision support. Automated sweetpotato disease inspection, however, remains challenging because field images exhibit substantial variations in leaf sc...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

👤 Thorsten Rudroff 📅 2026-10-29 📖 1 min 📝 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

Miles Wischnewski 2026-10-29 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu
2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

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Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

👤 Taomei Guo 📅 2026-10-31 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang
📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

Yange Wei
2026-10-29 1 min 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong 2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Sunday Daily Thread: What's everyone working on this week?

👤 /u/AutoModerator 📅 2026-11-01 📖 1 min 📝 211 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: What's Everyone Working On This Week? 🛠️</h1> Hello r/Python! It's time to share what you've been working on! Whether it's a work-in-progress, a completed masterpiece, or just a rough idea, let us know what you're up to!</p...

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https://www.reddit.com/r/Python/comments/1xkp2bx/sunday_daily_thread_whats_everyone_working_on/

AI text watermarking and quality loss

👤 danielfoster 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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An Italian farmer found some unknown seeds in his attic and planted them

👤 ecares 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Memdoor: A team of 5 Qwen3.8-27B models beats Fable 5 on LiveCodeBench Hard

👤 guregodevo 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Altoids Project

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When electric cars were for women

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Constructing a Quine from Kleene's Second Recursion Theorem

👤 mkckto 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

> **Summary:** Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

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Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

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VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

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http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

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A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

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Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

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ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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Guest Editorial: World Haptics Conference 2025

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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http://ieeexplore.ieee.org/document/11741002

Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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http://ieeexplore.ieee.org/document/11743154

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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http://ieeexplore.ieee.org/document/11740941

Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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http://ieeexplore.ieee.org/document/11740999

The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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http://ieeexplore.ieee.org/document/11741000

Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Plasma p-tau181, Aβ1-42, and GFAP across Alzheimer’s disease, mixed-type Alzheimer’s disease, mild cognitive impairment, anxiety disorders, and depressive disorders

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Haifeng Lou, Chenyun Huang, Ji Shen, Dake Wang

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Partial convergence of alternative polyadenylation changes across ALS-Associated RNA-Binding proteins

📖 1 min 📝 10 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Changhe Ji

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Genetic variants in NEF2L2 rs35652124 and GABRB3 rs2081648 genes: association with autism spectrum disorder in Iraqi children

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zainab Mohammed Burhan, Wisam Hindawi Hoidy

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Propofol suppresses intrinsic excitability and alters excitatory synaptic plasticity in ventral tegmental area dopamine neurons

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Min Hee Lee, Dong Kwan Kim

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https://www.sciencedirect.com/science/article/pii/S0306452226006986?dgcid=rss_sd_all

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

> **Summary:** CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

> **Summary:** CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard 2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

👤 Yoshihiro Watanabe 📅 2026-10-30 📖 1 min 📝 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Adaptive multiscale optical spatial filtering via a polarization-tunable phase-change micropore array

👤 Yikai Chen 📅 2026-10-30 📖 1 min 📝 62 words LOW VISION

Summary: As artificial intelligence and machine vision tasks evolve toward high throughput and low latency, all-optical analog signal processing demonstrates tremendous potential. However, the optical responses of most conventional static structures are fixed, rendering dynamic reconfiguration challenging. D...

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Visual acuity measurements in monochromatic and low-blue illumination

👤 Dragan Sekulovski 📅 2026-10-30 📖 1 min 📝 76 words LOW VISION

Summary: It is well known that visual acuity increases with light intensity. We report on the possibility to further increase visual acuity by changing the spectrum of the light while keeping light intensity fixed, or alternatively, provide the same visual acuity at lower light intensity. For a broad-spectru...

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Sensor-aware physical modelling for deep learning-based single-photon imaging

👤 Yu Pan 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Deep learning-based single-photon imaging commonly relies on synthetic time-resolved histograms with incompletely specified sensing physics. We present a sensor-aware framework that represents the source-receiver response, spatially varying background, detector artifacts, and first-detection acquisi...

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FDUNet-LRSA: efficient frequency-adaptive segmentation for automated crop disease inspection

Kai Huang 2026-10-31 1 min 56 words LOW VISION

Summary: The transition toward industrialized and automated agriculture requires reliable machine vision for high-throughput crop health monitoring and field decision support. Automated sweetpotato disease inspection, however, remains challenging because field images exhibit substantial variations in leaf sc...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

> **Summary:** CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

> **Summary:** The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

👤 Thorsten Rudroff 📅 2026-10-29 📖 1 min 📝 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

Miles Wischnewski 2026-10-29 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

Carlos M Gómez 2026-10-28 1 min 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang
📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

Yange Wei
2026-10-29 1 min 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong 2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42963327/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261031184445&v=2.20.1

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964054/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261031184445&v=2.20.1

Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964133/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261031184445&v=2.20.1

Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964233/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261031184445&v=2.20.1

Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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https://pubmed.ncbi.nlm.nih.gov/42964313/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261031184445&v=2.20.1

A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

TCP Use-After-Free: CVE-2026-98432 / CVSS 9.8

👤 oger 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://atlasum.net/blog/cve-2026-98432-linux-ipv4-tcp-uaf Comments URL: https://news.ycombinator.com/item?id=50319143 Points: 212 <p...

🔗 **Read full article:**
https://atlasum.net/blog/cve-2026-98432-linux-ipv4-tcp-uaf

Show HN: NES Contra (1987) port (ASM –> C and then C –> WebAssembly)

👤 tevlon 📅 2026-10-31 📖 1 min 📝 87 words HACKER NEWS

> **Summary:** Based on vermiceli's annotated disassembly of Contra: https://github.com/vermiceli/nes-contra-us. Thanks for the great work. I ported it in two steps. First to C: the game logic is translated routine by routine from the 650...

🔗 **Read full article:**
https://epoyraz.github.io/nes-contra-us/

Anthropic blocks users for anything touching training larger models

👤 pretext 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/jiahaoshao1/status/2116475922190922088 Comments URL: https://news.ycombinator.com/item?id=50319358 Points: 95 ...

🔗 **Read full article:**
https://twitter.com/jiahaoshao1/status/2116475922190922088

HackerTube: YouTube built from videos recommended by HN

👤 karpathy 📅 2026-10-31 📖 1 min 📝 25 words HACKER NEWS

> **Summary:** YouTube-style UI over 6,454 videos sourced from 39,834 HN comments since 2007. https://hackertube.org/ <hr /> Comments URL: https://news.ycombinator.com/item?id=50319510 <...

🔗 **Read full article:**
https://karpathy.bearblog.dev/hacker-tube/

Surviving "The Graduate"

👤 samizdis 📅 2026-10-31 📖 1 min 📝 14 words HACKER NEWS

> **Summary:** https://archive.ph/Eqn1F <hr /> Comments URL: https://news.ycombinator.com/item?id=50319563 Points: 28 # Comments: 3

🔗 **Read full article:**
https://www.newyorker.com/culture/the-weekend-essay/dustin-hoffman-on-surviving-the-graduate

With 86% of the words, you might understand only 38% of an anime's lines

👤 lakesare 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://learnfromx.com/research/anime-word-coverage-vs-lines-understood Comments URL: https://news.ycombinator.com/item?id=50320028...

🔗 **Read full article:**
https://learnfromx.com/research/anime-word-coverage-vs-lines-understood

Distributed Proofreaders

👤 gone35 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.pgdp.net/c/ Comments URL: https://news.ycombinator.com/item?id=50320137 Points: 11 # Comments: 1

🔗 **Read full article:**
https://www.pgdp.net/c/

CAG repeat length shapes bidirectional functional connectome alterations in spinocerebellar ataxia type 3: a multimodal MRI study

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Xianfei Yang, Haishuo Xia, Yuwei Hu, Can Xu, Xingang Wang, Qingxin Deng, Qian Xie, Limeng Dai, Qifang Yang, Changying Li, Li Gui, Chen Liu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007013?dgcid=rss_sd_all

Coarse motor-state structure across paired preoperative fMRI and intraoperative ECoG in patients with brain tumors

📖 1 min 📝 36 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jingxi Hu, Jie Ma, Wanwan Guo, Yang Jiao, Jiaqi Yang, Hongwei Wang, Yiheng Yin, Renke He, Gang Cheng, Hongmin Bai, Hairong Zheng, Dong Liang, Jianning Zhang, Zhanli Hu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006257?dgcid=rss_sd_all

Parietal structural and functional differentiation associated with visual tracking across motor participation profiles

📖 1 min 📝 26 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Keying Zhang, Meng Qi, Ling Jiang, Qing Yan, Yuyan Wang, Shuang Bai, Chunmei Cao, Dong Zhang, Siyuan Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006178?dgcid=rss_sd_all

Trisomy 21 Perturbs Oligodendrocyte Developmental States and Oligo-Neuron Communication in Human Forebrain Assembloids

👤 Zeldich, E., Campbell, N. B., Khanna, S. D., Kim, M. J., Kharitonova, E. K., Weidman, M. P., Headings, C., Chai, N. X., Chung, C., Chung, S. M., Field, G., Linetsky, M., Thunemann, M., Campbell, J. D., Medalla, M. D.

📅 2026-10-31 📖 1 min 📝 145 words BIORXIV NEUROSCIENCE

Summary: Intellectual disability is a hallmark of Down syndrome (DS), caused by the triplication of human chromosome 21. Abnormal neuronal connectivity and impaired myelination are consistently observed in DS, yet the underlying mechanisms remain poorly understood. Here, we used DS-derived isogenic induced p...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763166v1?rss=1

Levodopa induced dyskinesia severity is linked to distinct microstructural alterations in nigrostriatal and brainwide neural circuits

👤 Moshchin, M., Osting, S., Yakel, E., Schultz, R. J., Cheng, K. P., Davis, J. E., Luu, C. P., Trevathan, J. K., Lake, W. B., Hurley, S. A., Purcell, E. K., Bishop, C., Ludwig, K., Suminski, A. J.

📅 2026-10-31 📖 1 min 📝 145 words BIORXIV NEUROSCIENCE

Summary: Dopaminergic replacement therapy via L-DOPA is the major medical therapy for Parkinsons Disease. Long-term L-DOPA administration is known to cause dyskinetic movements, L-DOPA Induced Dyskinesia (LID). It remains unclear if the LID related, maladaptive changes in brain structure/function are localiz...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763167v1?rss=1

An expanded toolbox for genetic access to molecularly defined cortical intratelencephalic neuron subtypes

Russ, J. B., Zhang, J. S., Xu, X. H., Zhong, W., Liang, T., Qiu, Z., Kaushalya, S., Zhang, Y., Ke-Lind, P., Huang, Z. J.

2026-10-31 1 min 199 words BIORXIV NEUROSCIENCE

Summary: Among the multiple cortical glutamatergic neuron classes, intratelencephalic (IT) neurons represent the largest class and project within the cortex and striatum. IT neurons have been amplified and diversified throughout mammalian evolution and are thought to underlie increasingly complex mammalian c...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.762849v1?rss=1

State-dependent modulation of cross-area sensory processing in mouse neocortex

Han, S., Helmchen, F.

2026-10-31 1 min 205 words BIORXIV NEUROSCIENCE

Summary: Perception and decision-making rely on distributed cortical circuits that transform sensory inputs into choices and actions. Such cortical processing varies with behavioral state, such that the same inputs can be handled differently across states. However, how behavioral state shapes sensory process...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763148v1?rss=1

HERC1 Loss of Function Causes a Developmental Disorder with Impaired Hippocampal Function

Perez-Villegas, E. M., Garcia-Diez, M., Costa-Sastre, L., Hernandez-Curiel, J. M., Verdaguer-Farre, G., Sol-Cullere, P., Lopez-Murcia, F. J., Carrion, A. M., Ventura, F., Armengol, J. A., Rosa, J. L.

2026-10-31 1 min 265 words BIORXIV NEUROSCIENCE

Summary: Pathogenic variants in HERC1 are associated with a syndromic neurodevelopmental disorder characterized by macrocephaly, developmental delay, intellectual disability, and psychomotor dysfunction. Although genetic and molecular studies suggest a loss-of-function mechanism, definitive in vivo evidence ...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.763099v1?rss=1

An atlas of the aging glycoproteome reveals how glycosylation modulates autoantibody binding in the brain

Schoof, M., Reitenbach, J., Abraham, K., Tilton-Low, G., Wu, D., Chithrananda, S., Freund, Y., Channappa, D., Pepi, L. E., Dunn, J. E., Wernig, M., Han, M. H., Iram, T., Seeley, E. H., Guldner, I. H., Bertozzi, C. R., Wyss-Coray, T.

2026-10-31 1 min 180 words BIORXIV NEUROSCIENCE

Summary: N-glycosylation generates substantial molecular diversity, yet how glycan composition varies throughout aging remains poorly understood. We mapped the N-glycoproteome of brain, liver, spleen, kidney, and plasma across nine ages spanning the mouse lifespan, identifying 80,618 unique N-glycopeptides a...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.28.762617v1?rss=1

Changes in adaptive behavior, gut microbiota and fecal metabolites following an integrative intervention in children with autism spectrum disorder

📅 2026-10-31 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77828-4

Intrinsic motivation improves performance with less effort than extrinsic motivation

📅 2026-10-31 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44271-026-00550-4

Orexin and pupillary modulation for hunger control

👤 Seher KosarEva BraceyNikola GrujicYihui DuDaria Peleg-RaibsteinDenis Burdakovahttps://ror.org/02q2nz798ETH Zürich, Department of Health Sciences and Technology, Neuroscience Center Zürich, Institute for Neuroscience, Schwerzenbach 8603, Switzerland

📅 2026-10-28 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThis study reveals a genetically defined arousal-hunger circuit that enables mental engagement or external sensory input to naturally interrupt the urge to eat. These findings in mice suggest that...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619759123?af=R

Dlx1&2 directly promote expression of Cxcl14 to control synapse development and interneuron survival

👤 Teng GuoSusan LindtnerJia Sheng HuSiavash Fazel DarbandiYurong WangEmily Ling-Lin PaiAthéna R. YpsilantiCindy T. J. van VelthovenAlex S. NordJohn L. R. RubensteinaNina Ireland Laboratory of Developmental Neurobiology, Department of Psychiatry and Behavioral Sciences, University of California San Francisco, San Francisco, CA 94143bDepartment of Neurobiology, Physiology, and Behavior, Center for Neuroscience, University of California Davis, Davis, CA 95618cDepartment of Psychiatry and Behavioral Sciences, Center for Neuroscience, University of California Davis, Davis, CA 95618dhttps://ror.org/03cpe7c52Allen Institute for Brain Science, Seattle, WA 98109

📅 2026-10-28 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDlxgenes control multiple steps in the development of cortical interneurons (CIN), including synapse development and cell survival. Here we explore molecular mechanisms how theDlxgenes promote neo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2616055123?af=R

Lysine crotonylation acts as an epigenetic switch for glutamate neurotransmission and spatial memory

👤 Utsav MukherjeeBudhaditya BasuPratyush SuryavanshiStacy E. BeyerSaaman GhodsiNathan RobillardYann VanrobaeysEric B. TaylorJoseph GlykysTed AbelSnehajyoti Chatterjeeahttps://ror.org/036jqmy94Department of Neuroscience and Pharmacology, Carver College of Medicine, University of Iowa, Iowa City, IA 52242bhttps://ror.org/036jqmy94Iowa Neuroscience Institute, University of Iowa, Iowa City, IA 52242chttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Neuroscience, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Stead Family Department of Pediatrics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Genetics, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Fraternal Order of Eagles Diabetes Research Center, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Department of Molecular Physiology and Biophysics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242

📅 2026-10-28 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAlthough histone acetylation has long been considered a principal regulator of gene expression events underlying memory storage, the role of non-acetyl histone acyl modifications on long-term memo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2608460123?af=R

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Correction: Arts engagement, active inference, and allostatic regulation: a neurobiological framework for understanding the health effects of aesthetic experience

Pier Luigi Sacco 2026-10-13 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.2004139

Implementing the ILAE 2025 seizure taxonomy in medical software

Jukka Peltola 2026-10-13 1 min 241 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The International League Against Epilepsy (ILAE), working via a multidisciplinary task force, has issued a position paper with updated seizure classification guidelines (2025) replacing its previous guidance (2017), establishing a revised de-facto standard for clinical assessment and classification ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1877135

Cross-modal alignment with language tokens for unified multilingual image-to-text generation

Guanjun Lin 2026-10-14 1 min 274 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Current multimodal translation systems typically employ a cascaded pipeline where image captioning and machine translation are performed sequentially, leading to error propagation and semantic drift. This paper introduces a unified neural architecture that integrates visual encoding, English caption...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1826455

Correction: Post-stroke spasticity: advances in network mechanisms and therapeutics

Zhiqi Lin 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2035182

Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1920739

Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1960197

Single-cell dataset of juvenile mouse neurons from the thalamic reticular nucleus: whole-cell patch-clamp electrophysiology

Henry Markram 2026-10-29 1 min 216 words FRONTIERS NEUROSCIENCE

Summary: This dataset provides single-cell whole-cell patch-clamp electrophysiology for 19 interneurons of the thalamic reticular nucleus (TRN) in juvenile mice. The recordings were obtained from male C57Bl/6J mice sampled within the developmental window of postnatal days 24 to 34. Acute horizontal brain sli...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1927631

Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo
2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb442

Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

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CORRIGENDUM: In silico development and validation of Bayesian methods for optimizing deep brain stimulation parameters (2023 J. Neural Eng. 20 036015)

👤 Sumedh S Nagrale, Ali Yousefi, Theoden I Netoff and Alik S Widge

📅 2026-10-29 📖 1 min

📝 0 words JOURNAL NEURAL ENGINEERING

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32d

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

👤 Kwadwo Owusu Akuffo 📅 2026-10-30 📖 1 min 📝 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard 2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe 2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Adaptive multiscale optical spatial filtering via a polarization-tunable phase-change micropore array

Yikai Chen 2026-10-30 1 min 62 words LOW VISION

Summary: As artificial intelligence and machine vision tasks evolve toward high throughput and low latency, all-optical analog signal processing demonstrates tremendous potential. However, the optical responses of most conventional static structures are fixed, rendering dynamic reconfiguration challenging. D...

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Visual acuity measurements in monochromatic and low-blue illumination

Dragan Sekulovski 2026-10-30 1 min 76 words LOW VISION

Summary: It is well known that visual acuity increases with light intensity. We report on the possibility to further increase visual acuity by changing the spectrum of the light while keeping light intensity fixed, or alternatively, provide the same visual acuity at lower light intensity. For a broad-spectru...

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Sensor-aware physical modelling for deep learning-based single-photon imaging

👤 Yu Pan 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Deep learning-based single-photon imaging commonly relies on synthetic time-resolved histograms with incompletely specified sensing physics. We present a sensor-aware framework that represents the source-receiver response, spatially varying background, detector artifacts, and first-detection acquisi...

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FDUNet-LRSA: efficient frequency-adaptive segmentation for automated crop disease inspection

👤 Kai Huang 📅 2026-10-31 📖 1 min 📝 56 words LOW VISION

Summary: The transition toward industrialized and automated agriculture requires reliable machine vision for high-throughput crop health monitoring and field decision support. Automated sweetpotato disease inspection, however, remains challenging because field images exhibit substantial variations in leaf sc...

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Emerging light-driven neuromorphic hardware for artificial intelligence

Dimitris Tsoukalas 2026-10-31 1 min 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

👤 Thorsten Rudroff 📅 2026-10-29 📖 1 min 📝 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

Miles Wischnewski 2026-10-29 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

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Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

👤 Taomei Guo 📅 2026-10-31 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang
📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

👤 Yange Wei
📅 2026-10-29 📖 1 min 📝 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong 2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

👤 Blandina T Mmbaga 📅 2026-10-31 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

👤 Matias Valdenegro-Toro 📅 2026-10-31 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

📅 2026-05-18 📖 1 min 📝 183 words DREAMING

> **Summary:** The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

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Hackers hijack Egypt highway billboard and Malaysian screens to show porn

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UK Green party drops longstanding opposition to nuclear power

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> **Summary:** Article URL: https://www.theguardian.com/politics/2026/oct/05/green-party-nuclear-power-policy Comments URL: <a href="https://news.ycombinator.com/item?id=50318189">https://news.ycombinator.com/...

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The Impact of Generative AI on Critical Thinking: Self-Reported (2025)

👤 the-mitr 📅 2026-10-31 📖 1 min 📝 14 words HACKER NEWS

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It's the Memory Latency, Stupid

👤 ntur1337 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Shaun Thomas: Burnt Toast

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GPT-5.6 Sol can beat Factorio on peaceful mode

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The Duty of a Scholar

👤 gslin 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

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How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

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Visuospatial attention modulates audiovisual temporal perception: Context-dependent associations between prestimulus alpha lateralization and perceptual reports

1 min 30 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Zeliang Jiang, Yuqing Zhao, Weicong Ren, Jie Zhang, Xinyuan Liu, Fei Yu, Xingwei An, Wenxiao Zhong, Di Wu, Pan Zhang, Zhijie Zhang

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The visual cortex and beyond: investigating long-term cortical plasticity following monocular deprivation

📖 1 min 📝 29 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Franziska Wagner, Lukas Grajewski, Alexander Schmidt, Ulrike Teschner, Markus Ullsperger, Kathrin Finke, Christian Gaser, Robert Dahnke, Carsten M. Klingner, Stefan Brodoehl

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Distinct neural patterns of valence and arousal in the developing brain: A representational similarity analysis of MEG

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Ioannis Ntoumanis, Christos Papadelis

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Phenotypic screening in ALS motor neurons identifies SIRT1 activation as a genotype-independent neuroprotective strategy

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Can neuroimaging unlock precision therapeutics in psychosis spectrum disorders?

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HERC2–UBE3A structure uncovers a domain-motif protein network potentially altered in Angelman and related syndromes

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Post-stroke cognitive impairment in an Alzheimer’s disease mouse model

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Numerals in news headlines act as an information-gateway signal, increasing engagement, perceived informativeness, and memory

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Visual tagging frequency influences SSVEP and early ERP responses during detection and working memory

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Aging alters the distribution, stability, and transcriptional signature of engram cells

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Sustained antidepressant-like effects of psychedelics through 5-HT_{2A}R–β-arrestin signaling

👤 Sheng Wang 📅 2026-10-23 📖 1 min 📝 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02497-5 This study shows that the sustained antidepressant-like effect of psychedelics is driven by 5-HT2A receptor–β-arrestin signaling. Repeated low...

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Identification of candidate markers for the isolation of neuron-specific extracellular vesicles

David R. Walt | 2026-10-23 | 1 min | 41 words | NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02466-y The authors developed methods to identify extracellular vesicles (EVs) in the blood that come from the brain by capturing EV surface proteins....

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Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao | 2026-10-02 | 1 min | 354 words | FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

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https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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A systematic review of machine learning approaches for MRI-based brain tumor analysis

👤 Almira Melsovna Kustubayeva
📅 2026-10-12 📖 1 min 📝 213 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Machine learning is now widely applied to magnetic resonance imaging-based brain tumor analysis, with task-specific performance metrics: the Dice similarity coefficient (DSC) quantifies segmentation overlap, while the area under the curve (AUC) and accuracy measure classification performance. A PRIS...

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Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

👤 Patrick Krauss
📅 2026-10-20 📖 1 min 📝 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer’s disease analysis

Arshad Ali 2026-10-29 1 min 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer’s disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

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Virtual brain models: a roadmap towards effective brain digital twins

Viktor Jirsa 2026-10-29 1 min 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-10-29 1 min 314 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMotor imagery (MI) can support self-initiated control, whereas brain–computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspended...

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Resolving the apparent paradox of multisensory expertise: coupling priors, common-cause priors, and how to distinguish among them

Pier Luigi Sacco 2026-10-29 1 min 348 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perceptual expertise is widely assumed to refine multisensory perception, yet the empirical literature appears inconsistent. Trained musicians show narrower audiovisual temporal integration windows and quicker segregation, whereas experienced wine tasters seem more influenced by a wine's color in th...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1960361

Editorial: Emerging talents in human neuroscience: psychophysiology and cognitive neuroscience 2025

Gaia Risso 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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Simulation-based neural dynamics of stress-related negative cognitive bias in depressive symptoms: a dynamical EEG/ERP framework

Jianping Liu 2026-10-29 1 min 192 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis simulation-based computational study develops a non-linear state-space framework for stress-related negative cognitive bias and multivariate EEG/ERP observations.MethodsA three-state model represents negative bias, cognitive control, and arousal, while symptom load and normalized mo...

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Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

👤 Sivakumar Balasubramanian
📅 2026-10-20 📖 1 min 📝 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

👤 Lijuan Chen
📅 2026-10-29 📖 1 min 📝 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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Case Report: Probable paroxysmal sympathetic hyperactivity during vascular and postoperative critical illness with bilateral white matter abnormalities

Lei Wang 2026-10-29 1 min 194 words FRONTIERS NEUROSCIENCE

Summary: BackgroundRecognition of paroxysmal sympathetic hyperactivity (PSH) during vascular and postoperative critical illness is complicated by overlapping infection and seizure features.Case presentationA 37-year-old man developed encephalopathy and rigidity after extensive arterial thrombosis, thrombecto...

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A systematic review and meta-analysis of quantitative pain outcomes in degenerative thoracic myelopathy

Oliver D. Mowforth 2026-10-29 1 min 322 words FRONTIERS NEUROSCIENCE

Summary: IntroductionDegenerative thoracic myelopathy (DTM) refers to symptomatic compression of the thoracic spinal cord secondary to degenerative pathology, including thoracic disc herniation (TDH), ossification of the ligamentum flavum (OLF) and ossification of the posterior longitudinal ligament (OPLL). ...

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From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard 2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe 2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Adaptive multiscale optical spatial filtering via a polarization-tunable phase-change micropore array

Yikai Chen 2026-10-30 1 min 62 words LOW VISION

Summary: As artificial intelligence and machine vision tasks evolve toward high throughput and low latency, all-optical analog signal processing demonstrates tremendous potential. However, the optical responses of most conventional static structures are fixed, rendering dynamic reconfiguration challenging. D...

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Visual acuity measurements in monochromatic and low-blue illumination

Dragan Sekulovski 2026-10-30 1 min 76 words LOW VISION

Summary: It is well known that visual acuity increases with light intensity. We report on the possibility to further increase visual acuity by changing the spectrum of the light while keeping light intensity fixed, or alternatively, provide the same visual acuity at lower light intensity. For a broad-spectru...

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Sensor-aware physical modelling for deep learning-based single-photon imaging

👤 Yu Pan 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Deep learning-based single-photon imaging commonly relies on synthetic time-resolved histograms with incompletely specified sensing physics. We present a sensor-aware framework that represents the source-receiver response, spatially varying background, detector artifacts, and first-detection acquisi...

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FDUNet-LRSA: efficient frequency-adaptive segmentation for automated crop disease inspection

👤 Kai Huang 📅 2026-10-31 📖 1 min 📝 56 words LOW VISION

Summary: The transition toward industrialized and automated agriculture requires reliable machine vision for high-throughput crop health monitoring and field decision support. Automated sweetpotato disease inspection, however, remains challenging because field images exhibit substantial variations in leaf sc...

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Emerging light-driven neuromorphic hardware for artificial intelligence

👤 Dimitris Tsoukalas 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: Optoelectronic neuromorphic systems have emerged as a promising hardware paradigm for next-generation artificial intelligence, combining the high bandwidth, parallelism, and wavelength selectivity of photonics with the adaptive plasticity of electronic materials. This mini-review provides a focused ...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux 📅 2026-07-18 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

👤 Thorsten Rudroff 📅 2026-10-29 📖 1 min 📝 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

Miles Wischnewski 2026-10-29 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan
📅 2026-10-30 📖 1 min 📝 33 words

TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

👤 Taomei Guo 📅 2026-10-31 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang
📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

👤 Yange Wei
📅 2026-10-29 📖 1 min 📝 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong 2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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SMS Tier-Encoding Framework

/u/Acid_Crows | 2026-10-31 | 5 min | 1066 words | REDDIT PYTHON

Summary: <!-- SC_OFF --> <h1>An algebraic notation, clean reference implementation, and measured performance benchmark for managing extremely large or highly varied magnitudes before exact processing.</h1> <h1>Core Architecture</h1> <h1>• The Notation: Built on the SMS dynamic ...

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https://www.reddit.com/r/Python/comments/1xk9tmf/sms_tierencoding_framework/

Resource Sharing: Python Recursion Visualizer

/u/Sea-Ad7805 | 2026-10-31 | 1 min | 93 words | REDDIT PYTHON

Summary: <!-- SC_OFF --> Programming using recursion can be powerful, but beginners often struggle with recursion. Resource invocation_tree, a Python recursion visualizer, can help. See this <a href="https://www.invocation-tree.com/#codeurl=https://raw.githubusercont...

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https://www.reddit.com/r/Python/comments/1xka67h/resource_sharing_python_recursion_visualizer/

Pushing ThreadingHTTPServer past toy use: long-lived streams, a semaphore cap, and when to give up a

/u/ lovespeedtest | 2026-10-31 | 2 min | 599 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I've been running a small home media server for about a year, written entirely with the standard library: <code>http.server.ThreadingHTTPServer</code>, <code>http.client</code>, <code>urllib</code>, <code>threading</code>, <code>queue</code>. No framework and no thi...

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https://www.reddit.com/r/Python/comments/1xk4n1t/pushing_threadinghttpserver_past_toy_use/

Digital minimalism

wilbertliu | 2026-10-31 | 1 min | 13 words | HACKER NEWS

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Serbia just elected the world's brainiest parliament

👤 input_sh 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Molly(Signal fork) has not been updated since 3 months and is vulnerable

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We Took over ChatGPT Accounts

👤 armcat 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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Prospero: A new coding harness for a more civilized age

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Why Did Meta Suspend a US Senate Candidate's Instagram in the Middle of a Rally?

👤 ck2 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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AI startup hacks Anthropic, gains access to employee details, internal messages

👤 healsdata 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.indiatoday.in/technology/news/story/ai-startup-hacks-anthropic-in-12-hours-gains-access-to-employee-details-and-internal-messages-3025452-2026-10-26">https://www.indiatoday.in/technology/news/story/ai-startup-hacks-anthropic-in-12-hours-gains-access-to-employee-d...

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Retired professor arrested for blocking delivery robot. Will do it again

👤 dxs 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

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https://www.cnn.com/2026/10/31/us/philadelphia-woman-delivery-robots-cec

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

👤 Adriel Carridice 📅 2025-11-04 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Single-cell RNA sequencing reveals vascular heterogeneity and cell–cell communication remodeling in the glioma blood–tumor barrier

📖 1 min 📝 31 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Zhipeng Zhao, Jiachen Lang, Haiyan Gu, Luodan Wang, Jianchang Luo, Jiawang Lang, Qiongying Yao, Xinyu Qiao, Zida Wang, Ling Yin, Caidi He

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004579?dgcid=rss_sd_all

Flavonoid chronotherapy normalizes circadian dynamics and sleep architecture in mice with painful nerve injury

👤 Launonen, H., Dai, W., Wen, M., Milicevic, N., Koskinen, M.-K., Pertovaara, A., Kalso, E., Wigren, H.-K., Palada, V.

📅 2026-10-30 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: Neuropathic pain (NP) is linked to disrupted sleep and circadian abnormalities upon painful peripheral nerve injury. By performing spared nerve injury (SNI) in circadian reporter PER2::LUC mice, we characterized the real-time changes in rhythmicity of the core clock gene Per2 in the pain-regulating ...

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Tau-CTFP: A connectome-informed continuous-time first-passage model for forecasting the onset of regional tau PET abnormality

👤 Aslay, C., Gispert, J. D., Villoslada, P., Alzheimer's Disease Neuroimaging Initiative

📅 2026-10-30

📖 1 min 📝 247 words BIORXIV NEUROSCIENCE

Summary: Tau PET burden is associated with subsequent cognitive decline in Alzheimer's disease, motivating forecasts for monitoring and studies of spatially targeted therapies. We introduce Tau-CTFP, a connectome-informed generative continuous-time first-passage model for first observed regional onsets of ta...

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Longitudinal persistence of EEG brain fingerprints:phase and complex coherency geometry across sessions andfunctional states

👤 Goldstein, D., Deeb, A., Nassir, B., Sawaed, S., Zaidenberg, N. J.

📅 2026-10-30 📖 1 min

📝 222 words

BIORXIV NEUROSCIENCE

Summary: EEG connectivity can distinguish individuals, but the persistence of its phase and magnitude structure across visits and functional states remains incompletely understood. We examined two geometric representations of complex coherency: unit phasors and full complex coherency, alongside triangle holo...

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Chronic unpredictable stress induces negative cognitive bias in rats with sex-specific changes in neurogenesis and functional connectivity

👤 Namchuk, A. B., Hodges, T., McGovern, A., Galea, L.

📅 2026-10-31 📖 1 min 📝 262 words

BIORXIV NEUROSCIENCE

Summary: Background: Females are more likely to be diagnosed with major depressive disorder (MDD) than males. Cognitive symptoms of MDD include negative cognitive bias, the perception of ambiguous situations as negative, which predicts MDD episode onset and severity. MDD is associated with changes to functio...

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Fixel-wise Differences in White Matter Fiber-bundle Cross-section identified in Children who Stutter

👤 Weerathunge, H., Liu, Y., Hobler, F., Angstadt, M., Jahn, A., Chow, H. M., Chang, S.-E.

📅 2026-10-31

📖 1 min 📝 269 words

BIORXIV NEUROSCIENCE

Summary: Developmental stuttering has been repeatedly linked to decreased white matter integrity and structural connectivity differences in major white matter tracts. Prior diffusion-weighted MRI studies reporting such white matter structural differences involved voxel-averaged tensor measures (e.g., fractio...

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Decomposing Parts of Speech: Neural Encoding of Semantic and Grammatical Dimensions and Their Combinational Structure

👤 Huang, Z., Bekemeier, N., Hervais-Adelman, A.

📅 2026-10-30 📖 1 min 📝 242 words

BIORXIV NEUROSCIENCE

Summary: Conventional part-of-speech (PoS) labels such as nouns, verbs and adjectives conflate lexical-semantic roots, grammatical functions, and their combinations, making it unclear whether neural sensitivity to PoS reflects either dimension or the way they are combined. We adopted Croft's typological fram...

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Full-Term and Preterm Differences in Human Brain Functional Connectivity Development Across Birth

👤 Ji, L., Cotter, D., Banker, S., Trentacosta, C. J., Thomason, M. E.

📅 2026-10-30 📖 1 min 📝 203 words

BIORXIV NEUROSCIENCE

Summary: We possess a limited understanding of how preterm and full-term brains develop during the earliest stages of life, from the fetal to neonatal period, and how they respond to the birth transition. Here, we characterize brain resting-state functional connectivity (RSFC) using over 1000 longitudinal fe...

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From Images and Words to Modality-Independent Semantic Representations: When and Where They Emerge

Piron, N., Desbordes, T., Olasagasti, I. M., Schwartz, S., Kazanina, N.

2026-10-30 1 min

240 words

BIORXIV NEUROSCIENCE

Summary: Conceptual knowledge can be accessed from different modalities, such as seeing an image or reading a written word. While prior neuroimaging studies implicate anterior temporal cortex in modality-independent semantic representations, when and how representations evoked by images and words converge on...

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Connectivity predicts where computation forms and where it can be changed

Sale, C., Coraggio, M., Kaplan, D. M., Richardson, M. J.

2026-10-30 1 min 146 words

BIORXIV NEUROSCIENCE

Summary: How a fixed neural circuit supports many computations is a central question in neuroscience and artificial intelligence. Population dynamics show how computations are implemented, but not how connectivity constrains them. Here we show that recurrent connectivity contains a mesoscale organization bet...

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Multimodal profiling and intersectional genetics reveal CA1 border inhibitory neuron diversity and postnatal emergence of retrosplenial connectivity

👤 Martinez Vizcaino, N., Thrupp, N., Lorente-Echeverria, B., Daaboul, D., Alabi, M., Milian Alastruey, C., Linaro, D., Fourne, Y., Vandenbempt, J., Lukacsovich, D., Földy, C., Lamote, J., Wojno, M., Poovathingal, S., Fiers, M., Wierda, K., Lim, L., Calafate, S., De Wit, J.

📅 2026-10-31 📖 1 min 📝 168 words BIORXIV NEUROSCIENCE

Summary: Inhibitory neurons (INs) at the stratum radiatum/stratum lacunosum-moleculare border of hippocampal CA1 comprise diverse populations, including long-range-projecting neurons that innervate the retrosplenial cortex (RSCtx). However, their molecular organization and developmental connectivity remain p...

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Assessing the reliability of accelerated quantitative susceptibility mapping sequences in memory clinic patients

📅 2026-10-31 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Progressive decline in EEG low frequency spectral power during overnight sleep

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https://www.nature.com/articles/s41598-026-77865-z

TGF-β-induced ganglioside remodeling highlights GD2 as a dynamically regulated target in glioblastoma

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Chronic exercise induces a pathway-dependent electrophysiological signature in the mouse hippocampus

📅 2026-10-31 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s42003-026-11267-9

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne 2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

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Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka 2026-10-23 1 min 318 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca 2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

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Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

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A thalamus-centered network model of restless legs syndrome: an integrative hypothesis illustrated by three clinical phenotypes

Szabolcs Várbíró 2026-10-23 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Restless legs syndrome (RLS) is a clinically heterogeneous sensorimotor disorder whose pathophysiology reflects interacting biological mechanisms rather than a single causal pathway. Established influences—including brain iron biology, dopaminergic regulation, glutamatergic and GABAergic balance, ad...

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A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

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Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Learning Dynamic Affinity Diffusion for Multi-Contrast MRI Super-Resolution via Implicit Neural Representations

Bingding Huang 2026-10-29 1 min 61 words LOW VISION

Summary: Magnetic resonance imaging (MRI) super-resolution (SR) focuses on reconstructing high-quality, high-resolution target images from sparsely acquired signals while enhancing reconstruction performance through guidance from a reference image. In clinical practice, MRI SR contributes to reduced scan dur...

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Myopia prevalence among students in nine-year and independent compulsory education schools: a retrospective cross-sectional study

Zixuan Xiao
2026-10-30 1 min 66 words LOW VISION

Summary: CONCLUSION: The prevalence of myopia was relatively high and surged from low-grade to high-grade levels among compulsory students in the Nanshan district. An association between the educational transition model and myopia prevalence was observed. However, this association was exploratory, because th...

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Artificial intelligence for medication recognition and safety: a systematic review of detection, classification, OCR, and digital health applications

J Karthikeyan 2026-10-30 1 min 54 words LOW VISION

Summary: CONCLUSION: AI-based medication recognition is evolving from a computer vision task into a digital informatics component supporting medication safety, accessibility, verification, and patient care. Future research should prioritize diverse datasets, leakage control, independent multicenter validatio...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard
2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe
2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Adaptive multiscale optical spatial filtering via a polarization-tunable phase-change micropore array

Yikai Chen 2026-10-30 1 min 62 words LOW VISION

Summary: As artificial intelligence and machine vision tasks evolve toward high throughput and low latency, all-optical analog signal processing demonstrates tremendous potential. However, the optical responses of most conventional static structures are fixed, rendering dynamic reconfiguration challenging. D...

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Visual acuity measurements in monochromatic and low-blue illumination

Dragan Sekulovski 2026-10-30 1 min 76 words LOW VISION

Summary: It is well known that visual acuity increases with light intensity. We report on the possibility to further increase visual acuity by changing the spectrum of the light while keeping light intensity fixed, or alternatively, provide the same visual acuity at lower light intensity. For a broad-spectru...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang 📅 2026-10-28 📖 1 min 📝 72 words TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

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Joint a priori kinetic modeling with neural ODE for dual-tracer dynamic PET separation

👤 Su Ruan 📅 2026-10-28 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: Simultaneous dual-tracer dynamic PET imaging acquires a time series of images and could streamline oncology workflows by enabling the quantification of complementary biological processes in a single session. However, separating the superimposed tracer signals from the resulting mixed image sequence ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

Thorsten Rudroff 2026-10-29 1 min 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

👤 Miles Wischnewski
📅 2026-10-29 📖 1 min 📝 36 words

TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

👤 Felix Blankenburg 📅 2026-10-30 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

👤 Hamdan Hamdan 📅 2026-10-30 📖 1 min 📝 33 words TDCS TACS TRNS

> **Summary:** CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dyadic fNIRS Hyperscanning: Experimental Paradigms and What They Reveal About Interbrain Synchrony

👤 Miguel Castelo-Branco 📅 2026-10-28 📖 1 min 📝 65 words FNIRS

> **Summary:** Functional near-infrared spectroscopy (fNIRS) hyperscanning enables the simultaneous measurement of the brain activities of various individuals performing shared or interactive tasks and estimating their interbrain synchrony (IBS). IBS is believed to reflect different mechanisms, including social bo...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang 📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

👤 Yange Wei
📅 2026-10-29 📖 1 min 📝 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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https://pubmed.ncbi.nlm.nih.gov/42962872/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261031044157&v=2.20.1

Beyond Static Representations: Learning Electroencephalogram Spatiotemporal Dynamics for Enhanced Emotion Recognition

Wei Wu 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalogram (EEG)-based emotion recognition plays a pivotal role in brain-computer interfaces and mental health monitoring. However, conventional approaches often neglect the intrinsic spatial correlations among EEG channels and fail to effectively model both global temporal dependencies an...

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A two-stage RAG-LLM pipeline for roadmapping neuroscience research in operational environments

Cara A Spencer 2026-10-29 1 min 32 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Neuroscientific research and applications in real world environments are often constrained by limited access, equipment limitations, and high costs. Consequently, robustly selecting constructs to focus effort on is a high-impact task.

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Brain-computer interface technology for chronic pain management

Nabi Rustamov 2026-10-29 1 min 54 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Chronic pain remains inadequately managed by pharmacological and interventional therapies, which carry risks including opioid-related complications and limited long-term efficacy. Brain-computer interfaces (BCI) have emerged as a non-pharmacological therapeutic approach, most commonly ...

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Attention-BrainVerse: A Self-Adaptive Hybrid BCI-VR Interaction Model Driven by Attention

Rui Yang 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) is an emerging interactive technology that, when integrated with virtual reality (VR), exhibits significant potential for diverse applications. However, widespread adoption of BCI-VR systems remain hindered by challenges such as poor usability, high hardware costs, and...

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Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

Jennifer Collinger 2026-10-29 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally occurring dis...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Is the surgeon or hospital a larger cost driver for lumbar fusion? A hierarchical Bayesian approach using time-driven activity-based costing

Ahilan Sivaganesan 2026-10-30 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: In cases of single-level lumbar fusions performed at a multihospital health system, variation in cost between surgeons was 30-times greater than variation between hospitals, although residual/unexplained variance was considerable. These hypothesis-generating findings might enhance physi...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Implicit learning of melodic structure: A role for pitch?

📅 2024-01-22 📖 1 min 📝 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

Summary: I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Tell HN: Scribd Breached

👤 x_____ 📅 2026-10-31 📖 1 min 📝 392 words HACKER NEWS

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https://news.ycombinator.com/item?id=50312756

Switch Lowering in GCC

👤 signa11 📅 2026-10-31 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://xoranth.net/gcc-switch/ Comments URL: https://news.ycombinator.com/item?id=50314085 Points: 4 # Comments: 0

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https://xoranth.net/gcc-switch/

There are 13 ways to analyse a replication study, but only one of them is coherent.

noreply@blogger.com (Daniel Lakens)
2026-05-03 6 min 1361 words

TWENTY PERCENT STATISTICIAN

Summary: Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

Adriel Carridice
2026-06-22 1 min 0 words

BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

👤 Adriel Carridice 📅 2026-09-23 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

State-specific dynamic functional connectivity disturbances in spinal cord injury-related neuropathic pain

1 min 29 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chuchu Sun, Haotian Xin, Ling Wang, Qian Chen, Yu Wang, Beining Yang, Xin Chen, Fang Li, Jie Lu, Nan Chen

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004658?dgcid=rss_sd_all

Rules of disengagement: orexin-arousal and dopamine-utility as separable axes

👤 Grujic, N., Ploerer, V. I., Hamann, M., Tesmer, A., Bracey, E., Peleg-Raibstein, D., Burdakov, D.

📅 2026-10-30 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: Staying on task, leaving it, and coming back to it are processes we experience daily. They depend on task utility and arousal, but the neurochemical processes behind this interplay are unknown. In mice performing a decision-making task with shifting reward size, we identified latent engagement state...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761642v1?rss=1

Heterogeneous self-couplings enable scale-invariant memory and cognitive flexibility without criticality

👤 Damiani, F., Di Antonio, G., Epifanio, J., Vinci, G. V., Galluzzi, A., Mattia, M.

📅 2026-10-30 📖 1 min 📝 156 words BIORXIV NEUROSCIENCE

Summary: The hierarchical organization of the brain underlies its ability to process information across multiple time scales and support flexible behavior through context-dependent computation. Here, we study a recurrent neural network model with heterogeneous self-interactions across nodes, capturing key fe...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.25.761657v1?rss=1

Quantitative spatial connectomics of liver reveals a partially exposed sympathetic network that integrates multiple cell types across the portal tract

👤 Noon, L. A., Casal, G., Soriano-Navarro, M., Cattin, A.-L., Camacho, R., Acosta Umanzor, C., Rojo Sanz, C., Leal Tassias, A., Fernandez-Rodriguez, J., Verboven, E., Gomez Sanchez, J. A., Darsinou, M., Power, A., Riccio, A., Andersson, E. R., Burden, J. J., Lloyd, A. C.

📅 2026-10-30 📖 1 min 📝 160 words BIORXIV NEUROSCIENCE

Summary: Peripheral tissues are innervated by sparse axonal networks that regulate homeostasis, regeneration and disease, yet their three-dimensional architecture and cellular targets remain largely unknown. Here, we present an EM workflow for quantitative 3D spatial connectomics to define these networks. We...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761888v1?rss=1

Estrogen remodels large-scale cortical dynamics to promote behavioral persistence

Asokan, M. M., Le, A., Tsin, D., Sirrs, L., Cramer, M., Oline, S. N., Rogers, F. D., Falkner, A. L.

2026-10-30 1 min 232 words BIORXIV NEUROSCIENCE

Summary: Sex-steroid hormones remodel the brain on slow timescales of hours to days to coordinate adaptive behaviors with changing bodily physiology. Yet how this slow remodeling reorganizes and constrains neural computations that govern behavior on a moment-to-moment timescale remains poorly understood, par...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.25.761718v1?rss=1

Author Correction: Hypothalamic Grb10 enhances leptin signalling and promotes weight loss

2026-10-30 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s42255-026-01672-z

Serotonin signaling enhances phagocytosis by brain macrophages through Htr2b and alleviates brain injury after stroke

Peiying Li 2026-10-28 1 min 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 28 October 2026; doi:10.1038/s41593-026-02482-y The serotonin receptor Htr2b activation enhances lysosome biogenesis and phagocytic clearance of debris to mitigate neuroinflammation and brai...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02482-y

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Improvement in acuity following motion adaptation: The role of information at different spatial frequencies

👤 Steven C Dakin 📅 2026-10-29 📖 1 min 📝 74 words LOW VISION

Summary: Prolonged exposure (adaptation) to blur, flicker or motion can transiently enhance visual acuity (VA). Such improvement could be linked to decreased contrast sensitivity (CS) for low spatial frequency (SF) image structure. Here, we determine if and why low SFs are required for such acuity improvemen...

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Safety of Multiple Sclerosis Treatments During Pregnancy: An Overview of the EU Regulatory Approach to Risk Management Planning

Hedvig Marie Egeland Nordeng
2026-10-29 1 min 51 words LOW VISION

Summary: CONCLUSION: Despite efforts to evaluate safety of MS medicines during pregnancy, recruitment and data collection remain challenging. Planning for use of multiple real-world data sources-including primary data collection, secondary use of electronic health records, and analyses of case reports-may he...

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Learning Dynamic Affinity Diffusion for Multi-Contrast MRI Super-Resolution via Implicit Neural Representations

Bingding Huang
2026-10-29 1 min 61 words LOW VISION

Summary: Magnetic resonance imaging (MRI) super-resolution (SR) focuses on reconstructing high-quality, high-resolution target images from sparsely acquired signals while enhancing reconstruction performance through guidance from a reference image. In clinical practice, MRI SR contributes to reduced scan dur...

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Myopia prevalence among students in nine-year and independent compulsory education schools: a retrospective cross-sectional study

Zixuan Xiao
2026-10-30 1 min 66 words LOW VISION

Summary: CONCLUSION: The prevalence of myopia was relatively high and surged from low-grade to high-grade levels among compulsory students in the Nanshan district. An association between the educational transition model and myopia prevalence was observed. However, this association was exploratory, because th...

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Artificial intelligence for medication recognition and safety: a systematic review of detection, classification, OCR, and digital health applications

J Karthikeyan 2026-10-30 1 min 54 words LOW VISION

Summary: CONCLUSION: AI-based medication recognition is evolving from a computer vision task into a digital informatics component supporting medication safety, accessibility, verification, and patient care. Future research should prioritize diverse datasets, leakage control, independent multicenter validatio...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard
2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe
2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

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Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

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mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang 📅 2026-10-28 📖 1 min 📝 72 words TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

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Joint a priori kinetic modeling with neural ODE for dual-tracer dynamic PET separation

👤 Su Ruan 📅 2026-10-28 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: Simultaneous dual-tracer dynamic PET imaging acquires a time series of images and could streamline oncology workflows by enabling the quantification of complementary biological processes in a single session. However, separating the superimposed tracer signals from the resulting mixed image sequence ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

Thorsten Rudroff 2026-10-29 1 min 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

👤 Miles Wischnewski
📅 2026-10-29 📖 1 min 📝 36 words

TDCS TACS TRNS

Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu
2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Brain activation changes during working memory training: Individual capacity effects on functional near-infrared spectroscopy measures

👤 Tali Bitan 📅 2026-10-28 📖 1 min 📝 66 words FNIRS

Summary: Working memory (WM) training has shown promise for cognitive enhancement, but the neural mechanisms underlying training effects and offline consolidation processes remain unclear. This study examined the correspondence between behavioral and neural changes during WM training and how individual diffe...

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Dyadic fNIRS Hyperscanning: Experimental Paradigms and What They Reveal About Interbrain Synchrony

👤 Miguel Castelo-Branco 📅 2026-10-28 📖 1 min 📝 65 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) hyperscanning enables the simultaneous measurement of the brain activities of various individuals performing shared or interactive tasks and estimating their interbrain synchrony (IBS). IBS is believed to reflect different mechanisms, including social bo...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang 📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

Yange Wei
2026-10-29 1 min 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

👤 Yan Song 📅 2026-10-28 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

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A synchronized dual-scale platform combining flexible micro-ECoG and skull-screw EEG reveals frequency-dependent cortex-to-skull spectral gain in awake mice

👤 Zhaohui Song
📅 2026-10-28 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is the most scalable window onto brain dynamics, yet how cortical activity is transformed before reaching extracranial electrodes has largely been inferred from forward models rather than directly measured. We built a dual-scale sensing platform in awake mice that record...

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Federated Spiking GANs with Renyi Differential Privacy for privacy-preserving synthetic EEG generation in brain-computer interfaces

Garima Bajwa
2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) data used in brain-computer and brain-machine interfaces (BCI/BMI) contain sensitive, user-specific information, creating privacy risks in consumer applications such as gaming, human augmentation, and virtual reality. Existing methods like encryption and anonymization ei...

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Beyond Static Representations: Learning Electroencephalogram Spatiotemporal Dynamics for Enhanced Emotion Recognition

Wei Wu 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalogram (EEG)-based emotion recognition plays a pivotal role in brain-computer interfaces and mental health monitoring. However, conventional approaches often neglect the intrinsic spatial correlations among EEG channels and fail to effectively model both global temporal dependencies an...

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A two-stage RAG-LLM pipeline for roadmapping neuroscience research in operational environments

Cara A Spencer 2026-10-29 1 min 32 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Neuroscientific research and applications in real world environments are often constrained by limited access, equipment limitations, and high costs. Consequently, robustly selecting constructs to focus effort on is a high-impact task.

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Brain-computer interface technology for chronic pain management

Nabi Rustamov 2026-10-29 1 min 54 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Chronic pain remains inadequately managed by pharmacological and interventional therapies, which carry risks including opioid-related complications and limited long-term efficacy. Brain-computer interfaces (BCI) have emerged as a non-pharmacological therapeutic approach, most commonly ...

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Attention-BrainVerse: A Self-Adaptive Hybrid BCI-VR Interaction Model Driven by Attention

Rui Yang 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) is an emerging interactive technology that, when integrated with virtual reality (VR), exhibits significant potential for diverse applications. However, widespread adoption of BCI-VR systems remain hindered by challenges such as poor usability, high hardware costs, and...

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Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

Jennifer Collinger 2026-10-29 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally occurring dis...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

2024-01-22 1 min 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Saturday Daily Thread: Resource Request and Sharing! Daily Thread

👤 /u/ AutoModerator 📅 2026-10-31 📖 1 min 📝 180 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Resource Request and Sharing 📚</h1> Stumbled upon a useful Python resource? Or are you looking for a guide on a specific topic? Welcome to the Resource Request and Sharing thread! <h2>How it Works:</h2> Request: Can't...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xjvh23/saturday_daily_thread_resource_request_and/

Bunny CDN Security Incident

👤 SondreDag 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://status.bunny.net/incidents/p6gf1by9kr7m Comments URL: https://news.ycombinator.com/item?id=50308697 Points: 200 # Comments: 40

🔗 **Read full article:**
https://status.bunny.net/incidents/p6gf1by9kr7m

A Pickpocket's Guide to New York

👤 prismatic 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.newyorker.com/magazine/2026/11/02/a-pickpockets-guide-to-new-york Comments URL: <a href="https://news.ycombinator.com/item?id=50308986">https://news.ycombinator.com/item?id=...

🔗 **Read full article:**
https://www.newyorker.com/magazine/2026/11/02/a-pickpockets-guide-to-new-york

Infinite Mac: BeOS

👤 ecliptik 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.persistent.info/2026/10/infinite-mac-beos.html Comments URL: https://news.ycombinator.com/item?id=50309029 Points: 105 ...

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https://blog.persistent.info/2026/10/infinite-mac-beos.html

Show HN: To Honor Sir Patrick Stewart, Enjoy the TNG Episode Guessing Game

👤 francisofascii 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tng.episodle.com/ Comments URL: https://news.ycombinator.com/item?id=50311686 Points: 49 # Comments: 15

🔗 **Read full article:**
https://tng.episodle.com/

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

Deidre Artis | 2026-09-11 | 1 min | 14 words | EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

1 min | 23 words | BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600466X?dgcid=rss_sd_all

Altered language network microstructure links with impaired language function and development in very preterm/very low birth weight born adults

📖 1 min 📝 31 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Lena Gambarte, Jil Wendt, Rebecca Hippen, Marcel Daamen, Henning Boecker, Peter Bartmann, Dieter Wolke, Benita Schmitz-Koep, Dennis M. Hedderich, Christian Sorg, Aurore Menegaux

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600621X?dgcid=rss_sd_all

Common phenomenal and neural substrate geometry in visual motion perception

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kallum Robinson, Ariel Zeleznikow-Johnston, Jiahao Wu, Yumiko Yoshimura, Naotsugu Tsuchiya

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006282?dgcid=rss_sd_all

Accumulation of FAM98B and tRNA-Splicing Ligase Complex Proteins in Neuronal Intranuclear Inclusions in Spinocerebellar ataxia type 4.

👤 Gandelman, M., Figueroa, K. P., Classen, A., Scoles, D. R., Pulst, S. M.

📅 2026-10-30 📖 1 min

📝 149 words

BIORXIV NEUROSCIENCE

Summary: Background and Objectives: Spinocerebellar ataxia type 4 (SCA4) is caused by a GGC-repeat expansion in ZFHX3 encoding an expanded polyglycine tract causing p62- and ubiquitin-positive neuronal intranuclear inclusions (NIIs). In multiple polyglycine diseases NIIs sequester tRNA-splicing ligase comple...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761951v1?rss=1

Coordinated orienting behavior emerges following unilateral optogenetic activation of Superior Colliculus GABAergic neurons

👤 Suyama, H., Lupashina, T., Hamon, M., Sibille, J., Teh, K. L., Liebert, K., Larkum, M. E., Sachdev, R., Kremkow, J.

📅 2026-10-30 📖 1 min 📝 186 words BIORXIV NEUROSCIENCE

Summary: Orienting coordinates movements of the eyes, whiskers, nose, and other effectors, but how the superior colliculus (SC) recruits these components over time remains unclear. We combined unilateral optogenetic activation of SC GABAergic neurons with high-speed tracking of facial movements and Neuropixe...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.761971v1?rss=1

Imbalance between RNA-binding proteins and the WAVE regulatory complex underlies the toxic gain-of-function effects of the CYFIP2 Arg87Cys variant in neurodevelopmental disorders

👤 Kim, U. S., Jun, Y., Kim, H., Yang, Y., Kang, M., Ma, R., Kang, H. R., Ju, S., Jeong, H., Kim, S. Y., Kim, Y., Oh, J. Y., Kim, E., Kim, K., Kang, H., Kim, J. Y., Han, K.

📅 2026-10-30 📖 1 min 📝 299 words BIORXIV NEUROSCIENCE

Summary: Pathogenic variants in cytoplasmic FMR1-interacting protein 2 (CYFIP2) cause neurodevelopmental disorders, with hotspot substitutions at arginine 87 (Arg87) associated with particularly severe symptoms. CYFIP2 functions both as a component of the WAVE (Wiskott-Aldrich syndrome protein family verprol...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762134v1?rss=1

Impact of heat and sex on embryonic brain growth in dragons with temperature induced sex-reversal (Pogona vitticeps, Lacertilia: Agamidae)

👤 Hoops, D., Ceddia, A., Peters, R., Whiteley, S. L., Collin, S. P., Crowe-Riddell, J.

📅 2026-10-30

📖 1 min 📝 250 words

BIORXIV NEUROSCIENCE

Summary: Climate warming can influence sex determination and embryonic growth in oviparous reptiles that rely on external temperatures during development. Warmer incubation temperatures can stimulate growth in ectotherms until oxygen supply can no longer keep pace with growth rate; at which point metabolical...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761761v1?rss=1

Blue light compromises cellular fate of intrinsically photosensitive M&uumlller glial cells in the developing chicken retina.

👤 Marchese, N. A., Rios, M. N., Yaneff, A., Davio, C., Guido, M. E.

📅 2026-10-30 📖 1 min

📝 269 words

BIORXIV NEUROSCIENCE

Summary: M&uumlller glial cells (MCs), the predominant glial population in the retina, interface with all neuronal types to form the structural and functional core of the tissue. Beyond their established homeostatic roles, MCs have emerged as promising endogenous neural stem cells with remarkable regener...

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Spatially resolved single-cell atlas reveals cortical glial reactivity associated with regions of neuronal growth programs in both hemispheres after stroke

👤 Kenwood, M., Betz, D., Guttierez-Kuri, E., Derebery, B., Rhoton, P., Palavicinni, J. P., Ramirez, D. M. O., Goldberg, M. P.

📅 2026-10-30 📖 1 min 📝 222 words BIORXIV NEUROSCIENCE

Summary: Post-stroke recovery relies on structural remodeling of neuronal circuits in both the spared ipsilesional and contralesional cortices. RNA sequencing studies have identified distinct states of glial reactivity after stroke, but how these states are spatially organized at the single-cell level in rel...

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https://www.biorxiv.org/content/10.64898/2026.10.26.761917v1?rss=1

Goat Milk-derived Extracellular Vesicles modulate glial responses in vitro and attenuate neuroinflammation in vivo

Santos-Coquillat, A., Lozano-Picazo, P., Iriarte-Urdiain, E., Yang, J., Ucakar, B., Mwema, A., Auquiere, M., des Rieux, A.

2026-10-30 1 min 219 words BIORXIV NEUROSCIENCE

Summary: Milk-derived extracellular vesicles (MEV) represent a naturally derived, potentially scalable platform for immunomodulation, yet their effects on central nervous system (CNS) glial inflammation at defined particle-to-cell ratios remain insufficiently characterized. Here, we investigated the immunomo...

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https://www.biorxiv.org/content/10.64898/2026.10.26.761897v1?rss=1

Neuroinflammation is an early pathological process in a mouse model of prodromal Parkinsons disease

Mercado, G., Stetzik, L., Schulz, E., Meyerdirk, L., Lindquist, A., Cohen, L., Escobar Galvis, M. L., Steiner, J. A., Moore, D. J., Wesson, D. J., Brundin, P.

2026-10-30 1 min 246 words BIORXIV NEUROSCIENCE

Summary: Approximately 90% of Parkinson's disease (PD) patients exhibit olfactory deficits, which often precedes motor symptoms by years, and are a robust predictor for PD and subsequent cognitive decline. Pathological changes in the PD olfactory system, particularly synuclein (Syn) pathology in the olfactor...

Read full article:
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Pre-stress functional brain connectivity is associated with subsequent anxiety during war-related stress

Boshnak, N., Danan, N., Keren, H.

2026-10-30 1 min 139 words

BIORXIV NEUROSCIENCE

Summary: Individuals exposed to major real-world stressors show markedly different psychological outcomes, but whether this heterogeneity is associated with pre-existing neural characteristics remains unclear because neuroimaging is rarely performed beforehand. We acquired resting-state fMRI in healthy adult...

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https://www.biorxiv.org/content/10.64898/2026.10.24.761100v1?rss=1

Persistent cerebrovascular injury after preeclampsia in the STOX1A mouse model: implications for prokineticin signaling

Younes, H., Sergent, F., Kyritsi, I., Miguel, L., Montigon, O., Vaiman, D., Mohamed, B., Barbier, E. L., Alfaidy, N., Marquette, C.

2026-10-30 1 min 261 words

BIORXIV NEUROSCIENCE

Summary: Preeclampsia (PE) is a pregnancy-related hypertensive disorder associated with neurological complications that may persist beyond pregnancy. Prokineticins (PROK1 and PROK2) and their receptors (PROKR1 and PROKR2) regulate angiogenesis and inflammation and may be involved to PE-associated cerebrovasc...

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Kaempferol attenuates secondary brain injury after traumatic brain injury in mice with reduced neuroinflammation

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-78325-4

Parkinson’s disease progression and fronto-temporal disconnection relate to identity of minor hallucinations

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41531-026-01556-9

Microglial NUS1 deficiency drives dopaminergic neurodegeneration through intercellular iron dysregulation in Parkinson’s disease models

Haigang Ren 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02499-3 Using cellular and mouse models, this study reports that deficiency of the Parkinson’s disease risk gene NUS1 in microglia disrupts intercellu...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02499-3

Chronic progressive demyelination induces a microglial state with lysosomal dysfunction

Mikael Simons 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02489-5 Acute myelin injury responses are well defined, but microglia adaptations to chronic damage remain unclear. Mouse models reveal a chronic demy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02489-5

Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

Olfat A. MalakLindsay GrayAmmar AlyRobin W. BallJessica Gustin-MurrayMyriam MoujahidineGuadalupe CarrascoAndrew RoskoEhud Y. IsacoffA. Pejmun Haghighiahttps://ror.org/050sv4x28Buck Institute for Research on Aging, Novato, CA 94945bhttps://ror.org/04b6nzv94Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115cInscopix Inc., Mountain View, CA 94043dhttps://ror.org/03taz7m60Leonard Davis School of Gerontology, University of Southern California, Los Angeles, CA 90089ehttps://ror.org/01pxwe438Department of Physiology, McGill University, Montreal, QC H3G1Y6, Canadafhttps://ror.org/01an7q238Helen Wills Neuroscience Institute, University of California, Berkeley, CA 94720ghttps://ror.org/01an7q238Department of Neuroscience, University of California, Berkeley, CA 94720hhttps://ror.org/01an7q238Department of Molecular and Cell Biology, University of California, Berkeley, CA 94720

📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

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https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

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https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong
2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

MTDANet: lightweight multi-scale temporal hierarchy and distance-regularized linear attention network for motor imagery brain–computer interfaces

Binbin Chen, Chen Deng, Huanmin Ge, Xinhua Su and Yanfei Shen
2026-10-26 1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor-imagery (MI) electroencephalography (EEG) offers a non-invasive route to brain–computer interaction, yet reliable single-trial decoding remains difficult because task-related patterns are weak, nonstationary, and distributed over multiple temporal scales. This study investigates whe...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb330

AOR: adaptive orthogonal regression for depression feature selection with incomplete EEG channels

👤 Wenjia Dong, Zhijian Gong, Xueyuan Xu, Ke Liu, Fulin Wei and Li Zhuo

📅 2026-10-26 📖 1 min

📝 272 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Depression induces severe mental and physical impairments, highlighting the clinical need for rapid and reliable detection. Electroencephalography (EEG) directly reflects neural electrical activity and serves as a promising tool for depression investigation. However, the inherent non-stat...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32f

Multimodal regulation of neurite extension by extracellular matrix composition and electrical stimulation on aligned collagen fiber scaffolds

👤 Madalynn J Thompson, Joanna Borowska-Fielding, Ivy A Murphy, Ying Zhang, Victor Rafuse and John P Frampton

📅 2026-10-26 📖 1 min 📝 279 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Peripheral nerve regeneration is regulated by structural, biochemical, and electrical cues, yet their combined effects on neurite outgrowth remain unclear. In this study, an aligned fibrous scaffold was developed to evaluate how extracellular matrix (ECM) composition and electrical stimul...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Improvement in acuity following motion adaptation: The role of information at different spatial frequencies

👤 Steven C Dakin 📅 2026-10-29 📖 1 min 📝 74 words LOW VISION

Summary: Prolonged exposure (adaptation) to blur, flicker or motion can transiently enhance visual acuity (VA). Such improvement could be linked to decreased contrast sensitivity (CS) for low spatial frequency (SF) image structure. Here, we determine if and why low SFs are required for such acuity improvemen...

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Safety of Multiple Sclerosis Treatments During Pregnancy: An Overview of the EU Regulatory Approach to Risk Management Planning

👤 Hedvig Marie Egeland Nordeng
📅 2026-10-29 📖 1 min 📝 51 words LOW VISION

Summary: CONCLUSION: Despite efforts to evaluate safety of MS medicines during pregnancy, recruitment and data collection remain challenging. Planning for use of multiple real-world data sources-including primary data collection, secondary use of electronic health records, and analyses of case reports-may he...

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Learning Dynamic Affinity Diffusion for Multi-Contrast MRI Super-Resolution via Implicit Neural Representations

👤 Bingding Huang
📅 2026-10-29 📖 1 min 📝 61 words LOW VISION

Summary: Magnetic resonance imaging (MRI) super-resolution (SR) focuses on reconstructing high-quality, high-resolution target images from sparsely acquired signals while enhancing reconstruction performance through guidance from a reference image. In clinical practice, MRI SR contributes to reduced scan dur...

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Myopia prevalence among students in nine-year and independent compulsory education schools: a retrospective cross-sectional study

👤 Zixuan Xiao
📅 2026-10-30 📖 1 min 📝 66 words

LOW VISION

Summary: CONCLUSION: The prevalence of myopia was relatively high and surged from low-grade to high-grade levels among compulsory students in the Nanshan district. An association between the educational transition model and myopia prevalence was observed. However, this association was exploratory, because th...

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Artificial intelligence for medication recognition and safety: a systematic review of detection, classification, OCR, and digital health applications

J Karthikeyan 2026-10-30 1 min 54 words LOW VISION

Summary: CONCLUSION: AI-based medication recognition is evolving from a computer vision task into a digital informatics component supporting medication safety, accessibility, verification, and patient care. Future research should prioritize diverse datasets, leakage control, independent multicenter validatio...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Reliability of Clinical Oculomotor Tests in Children and Young Adults Following Moderate-Severe Traumatic Brain Injury

Mathilde Chevignard 2026-10-30 1 min 35 words LOW VISION

Summary: CONCLUSION: In-person ratings of standardised clinical OM tests demonstrated almost perfect inter-rater agreement, supporting their use in clinical practice following msTBI. However, video-based assessments show low reliability and require careful consideration and potentially modified protocols.

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Dynamic Projection Mapping via Fast 3D Gaussian Splatting with Blue-Noise Stochastic Sampling

Yoshihiro Watanabe 2026-10-30 1 min 64 words LOW VISION

Summary: Dynamic projection mapping (DPM) enables spatially aligned projection onto moving objects, but requires low-latency rendering and accurate appearance reproduction. 3D Gaussian splatting (3DGS) has recently emerged as an image-based approach for reconstructing 3D scenes solely from photographs and re...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

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Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

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mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang 📅 2026-10-28 📖 1 min 📝 72 words TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

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Joint a priori kinetic modeling with neural ODE for dual-tracer dynamic PET separation

👤 Su Ruan 📅 2026-10-28 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: Simultaneous dual-tracer dynamic PET imaging acquires a time series of images and could streamline oncology workflows by enabling the quantification of complementary biological processes in a single session. However, separating the superimposed tracer signals from the resulting mixed image sequence ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

Thorsten Rudroff 2026-10-29 1 min 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

👤 Miles Wischnewski
📅 2026-10-29 📖 1 min 📝 36 words

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Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

👤 Zhikun Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Brain activation changes during working memory training: Individual capacity effects on functional near-infrared spectroscopy measures

👤 Tali Bitan 📅 2026-10-28 📖 1 min 📝 66 words FNIRS

Summary: Working memory (WM) training has shown promise for cognitive enhancement, but the neural mechanisms underlying training effects and offline consolidation processes remain unclear. This study examined the correspondence between behavioral and neural changes during WM training and how individual diffe...

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Dyadic fNIRS Hyperscanning: Experimental Paradigms and What They Reveal About Interbrain Synchrony

👤 Miguel Castelo-Branco 📅 2026-10-28 📖 1 min 📝 65 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) hyperscanning enables the simultaneous measurement of the brain activities of various individuals performing shared or interactive tasks and estimating their interbrain synchrony (IBS). IBS is believed to reflect different mechanisms, including social bo...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang 📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

Yange Wei
2026-10-29 1 min 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

👤 Yan Song 📅 2026-10-28 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

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A synchronized dual-scale platform combining flexible micro-ECoG and skull-screw EEG reveals frequency-dependent cortex-to-skull spectral gain in awake mice

👤 Zhaohui Song
📅 2026-10-28 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is the most scalable window onto brain dynamics, yet how cortical activity is transformed before reaching extracranial electrodes has largely been inferred from forward models rather than directly measured. We built a dual-scale sensing platform in awake mice that record...

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Federated Spiking GANs with Renyi Differential Privacy for privacy-preserving synthetic EEG generation in brain-computer interfaces

Garima Bajwa
2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) data used in brain-computer and brain-machine interfaces (BCI/BMI) contain sensitive, user-specific information, creating privacy risks in consumer applications such as gaming, human augmentation, and virtual reality. Existing methods like encryption and anonymization ei...

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Beyond Static Representations: Learning Electroencephalogram Spatiotemporal Dynamics for Enhanced Emotion Recognition

👤 Wei Wu 📅 2026-10-29 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalogram (EEG)-based emotion recognition plays a pivotal role in brain-computer interfaces and mental health monitoring. However, conventional approaches often neglect the intrinsic spatial correlations among EEG channels and fail to effectively model both global temporal dependencies an...

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A two-stage RAG-LLM pipeline for roadmapping neuroscience research in operational environments

👤 Cara A Spencer 📅 2026-10-29 📖 1 min 📝 32 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Neuroscientific research and applications in real world environments are often constrained by limited access, equipment limitations, and high costs. Consequently, robustly selecting constructs to focus effort on is a high-impact task.

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Brain-computer interface technology for chronic pain management

Nabi Rustamov 2026-10-29 1 min 54 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Chronic pain remains inadequately managed by pharmacological and interventional therapies, which carry risks including opioid-related complications and limited long-term efficacy. Brain-computer interfaces (BCI) have emerged as a non-pharmacological therapeutic approach, most commonly ...

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Attention-BrainVerse: A Self-Adaptive Hybrid BCI-VR Interaction Model Driven by Attention

Rui Yang 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) is an emerging interactive technology that, when integrated with virtual reality (VR), exhibits significant potential for diverse applications. However, widespread adoption of BCI-VR systems remain hindered by challenges such as poor usability, high hardware costs, and...

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Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

Jennifer Collinger 2026-10-29 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally occurring dis...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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http://doi.org/10.1037/neu0001096

The oldest ice disappears and the Stone Age emerges from it

👤 dxs 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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Jules Verne's Lost Novel

👤 dnetesn 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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An Arrogant Way of Killing (2016)

👤 firefax 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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Show HN: Automatic deduction of FlashAttention kernel from ~25 lines of SQL

👤 alxmrs 📅 2026-10-30 📖 1 min 📝 194 words HACKER NEWS

> **Summary:** In this demo, we automatically deduce a FlashAttention kernel with ~25 lines of SQL, which just expresses vanilla attention. This is possible due to three portable SQL extensions that we’re building: - xarray-sql: a library that efficiently represents petabyte scale array datasets in the cloud as...

🔗 **Read full article:**
https://github.com/xqlsystems/xarray-sql/blob/6debe2495dcb4e8fac20ab97df5b89baae2a31a6/benchmarks/ml/01_attention.py

Embraer's Atech aims to double in size by 2030 with drone traffic control

👤 dvaplima 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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Through Captain Jean-Luc Picard, Patrick Stewart taught us the measure of a man

👤 apress 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bostonglobe.com/2026/10/29/metro/patrick-stewart-picard-star-trek-the-next-generation/ Comments URL: <a href="https://news.ycombinator.com/item?id=50309...

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AO3 Down for Maintenance

👤 idlewords 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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https://www.otwstatus.org

Sanstun: A sans-IO, parallel-friendly WireGuard implementation in Rust

👤 wh33zle 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.firezone.dev/blog/introducing-sanstun Comments URL: https://news.ycombinator.com/item?id=50310398 Points: 23 # Comments: 1

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https://www.firezone.dev/blog/introducing-sanstun

Linode Security Incident

👤 lacoolj 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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https://status.linode.com/incidents/nt58zhx3pmgm

France is getting download tax for content providers (€0.055/GB)

👤 bko 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.resetera.com/threads/france-is-getting-download-tax-for-content-providers-%E2%82%AC0-055-gb-for-data-delivered-to-french-users.1675109/">https://www.resetera.com/threads/france-is-getting-download-tax-for-content-providers-%E2%82%AC0-055-gb-for-data-delivered-to-...

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Type S and M errors as a “rhetorical tool”

noreply@blogger.com (Daniel Lakens)
2025-09-28 17 min 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens)
2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

> **Summary:** You can access the recording h.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

> **Summary:** We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Do neuroimaging data have higher-order interactions?

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Rikkert Hindriks, Fernando Santos, Matteo Neri, Prejaas Tewarie

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006117?dgcid=rss_sd_all

Pregnancy History and Loss of Normal PSEN2 Function Impact Female Late-Life Cognitive Processing, Inhibitory Signaling, and Gliosis

👤 Fang, J. I., Lehmann, L., Prater, K. E., Barker-Haliski, M. 📅 2026-10-30 📖 1 min 📝 250 words

BIORXIV NEUROSCIENCE

Summary: Background. Alzheimer's disease (AD) is well-documented to be more progressive in affected females. It is thus critical to define how female-specific life experiences interact with AD-related risk factors to impact disease trajectory and functional burden. For example, pregnancy exerts dramatic infl...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762624v1?rss=1

Neural mechanisms for resilient sensory computation under metabolic stress

Castedo, S. H., Padamsey, Z., Rochefort, N. L., Monasson, R.

2026-10-30 1 min 78 words

BIORXIV NEUROSCIENCE

Summary: Food-restricted mice can solve visual discrimination tasks despite reduced synaptic currents and increased noise resulting from cellular changes in the primary visual cortex. Combining neural recordings and modeling, we study the population structure of the noise and show it is mainly concentrated a...

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https://www.biorxiv.org/content/10.64898/2026.10.29.761790v1?rss=1

Comparative Proteomics Reveals Cell-Type-Specific Heterogeneity of Human Neural Cell-Derived Extracellular Vesicles

Chen, X., Lo, C. H., Zeng, J.

2026-10-30 1 min 233 words

BIORXIV NEUROSCIENCE

Summary: Extracellular vesicles (EVs) are increasingly explored as biomarkers and as drug delivery vehicles for the brain, yet how EV cargo differs among neural cell types at baseline remains poorly defined. This gap limits both the interpretation of disease-associated EV changes and the rational choice of E...

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Evolutionary tuning of complexin-synaptotagmin fusion clamp governs regulated neurotransmitter release

👤 Wang, Y., Bose, D., Griffith, D., Dittman, J. S., Krishnakumar, S. S.

📅 2026-10-30 📖 1 min

📝 190 words

BIORXIV NEUROSCIENCE

Summary: Synaptic vesicles must be held in a fusion-ready but restrained state until calcium influx triggers release. Synaptotagmin-1 (Syt1) and Complexin (CPX) both contribute to this fusion clamp, yet removing either protein produces opposite effects in different species. Whether this reflects genuine dive...

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Rapid switching of in-phase and anti-phase interhemispheric rhythms within a single theta cycle: a mechanistic explanation

Kailasa, S., Jedrasiak-Cape, I., Booth, V., Avitabile, D., Ahmed, O. J.

2026-10-30 1 min

221 words

BIORXIV NEUROSCIENCE

Summary: The granular retrosplenial cortex (RSG) supports memory consolidation and spatial navigation. In vivo recordings of local field potential (LFP) from RSG reveal coexisting gamma (30-80 Hz) and spline (100-160 Hz) oscillations, coupled to distinct phases of the background theta ([~]8 Hz) rhythm. Remar...

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Adaptive tunable sensing system inspired by scorpions for dynamic vibration perception in unstructured environments

2026-10-30 1 min 0 words

NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41528-026-00655-0

Eye-state-dependent reorganization of EEG microstates in a pediatric transdiagnostic cohort

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https://www.nature.com/articles/s41598-026-78232-8

Inflammatory biomarkers, stroke severity, and 1-month outcomes in minor acute ischaemic stroke by small-vessel occlusion status

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Ancestry-specific and multi-ancestry genome-wide association studies of restless legs syndrome

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Intruding delta waves during REM sleep are associated with tau, amyloid, and forgetting in aging

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https://www.nature.com/articles/s41467-026-78425-9

Editorial: Women in interacting minds and brains II

👤 Anna Ilona Roberts 📅 2026-10-26 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2001078

Imaging findings and clinical outcomes in neonates with isolated subependymal echogenicity on cranial ultrasound: a retrospective cohort study

Xuehua Zhang
2026-10-28 1 min 255 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo describe clinical and imaging findings in neonates with isolated subependymal echogenicity on cranial ultrasound and examine associations with recorded mortality and cerebral palsy.MethodsThis retrospective single-center cohort included 317 neonates admitted between January 2015 and Dece...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1945002

Effects of non-invasive brain stimulation combined with exercise training on pain and physical function in knee osteoarthritis: a systematic review and meta-analysis

Zhaoxiang Meng
2026-10-28 1 min 345 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundNon-invasive brain stimulation (NIBS) combined with exercise training (ET) is a promising treatment for knee osteoarthritis (KOA), but its effects on pain and physical function lack systematic evaluation.PurposeTo evaluate the effects of NIBS combined with ET on pain and physical function ...

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-16 1 min 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Oculomotor rehabilitation after concussion in adults (ORCA): a feasibility study

Tatyana Mollayeva 2026-10-30 1 min 313 words FRONTIERS NEUROSCIENCE

Summary: BackgroundOculomotor deficits are reported in up to 85% of adults after concussion and may contribute to prolonged symptoms and delayed return to daily activities. Oculomotor rehabilitation has been successfully implemented in younger age groups, however evidence in adults is lacking. Virtual realit...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1898653

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Predictors of improvement in distance presenting visual acuity after refraction and reading rehabilitation at 1.5 M print size in older adults with low vision

Rosalia Antunes-Foschini 2026-10-29 1 min 23 words LOW VISION

Summary: CONCLUSIONS: Clinical, refractive, and demographic factors were associated with dpVA improvement after refraction and successful reading rehabilitation in older adults with low vision.

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An open-source standardized pipeline for equitable observations of interactive behavioral dynamics: Theory-driven measurement, analysis, and masking

Travis J Wiltshire 2026-10-29 1 min 69 words LOW VISION

Summary: Behavioral observation has a long-standing tradition in psychology. However, its reliance on manual coding makes observational methods costly and prone to biases that arise from the inherent limits of coding schemes and from the subjectivity of human raters. Behavioral observation research therefore...

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Improvement in acuity following motion adaptation: The role of information at different spatial frequencies

👤 Steven C Dakin
📅 2026-10-29 📖 1 min 📝 74 words

LOW VISION

Summary: Prolonged exposure (adaptation) to blur, flicker or motion can transiently enhance visual acuity (VA). Such improvement could be linked to decreased contrast sensitivity (CS) for low spatial frequency (SF) image structure. Here, we determine if and why low SFs are required for such acuity improvemen...

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Safety of Multiple Sclerosis Treatments During Pregnancy: An Overview of the EU Regulatory Approach to Risk Management Planning

👤 Hedvig Marie Egeland Nordeng
📅 2026-10-29 📖 1 min 📝 51 words

LOW VISION

Summary: CONCLUSION: Despite efforts to evaluate safety of MS medicines during pregnancy, recruitment and data collection remain challenging. Planning for use of multiple real-world data sources-including primary data collection, secondary use of electronic health records, and analyses of case reports-may he...

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Learning Dynamic Affinity Diffusion for Multi-Contrast MRI Super-Resolution via Implicit Neural Representations

Bingding Huang 2026-10-29 1 min 61 words LOW VISION

Summary: Magnetic resonance imaging (MRI) super-resolution (SR) focuses on reconstructing high-quality, high-resolution target images from sparsely acquired signals while enhancing reconstruction performance through guidance from a reference image. In clinical practice, MRI SR contributes to reduced scan dur...

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Myopia prevalence among students in nine-year and independent compulsory education schools: a retrospective cross-sectional study

Zixuan Xiao 2026-10-30 1 min 66 words LOW VISION

Summary: CONCLUSION: The prevalence of myopia was relatively high and surged from low-grade to high-grade levels among compulsory students in the Nanshan district. An association between the educational transition model and myopia prevalence was observed. However, this association was exploratory, because th...

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Artificial intelligence for medication recognition and safety: a systematic review of detection, classification, OCR, and digital health applications

J Karthikeyan 2026-10-30 1 min 54 words LOW VISION

Summary: CONCLUSION: AI-based medication recognition is evolving from a computer vision task into a digital informatics component supporting medication safety, accessibility, verification, and patient care. Future research should prioritize diverse datasets, leakage control, independent multicenter validatio...

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Artificial intelligence innovations in low vision rehabilitation: a systematic review of emerging technologies

Kwadwo Owusu Akuffo 2026-10-30 1 min 51 words LOW VISION

Summary: CONCLUSION: AI assistive technologies for low vision demonstrate technical feasibility with promising performance. However, important evidence gaps limit clinical application: the absence of prospective validation, limited comparative effectiveness data, and minimal patient-centered outcomes. Future...

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Motor memory consolidation during sleep after stroke: the rehabilitation timing hypothesis

Faisal Hamed Aljamea 2026-10-30 1 min 53 words LOW VISION

Summary: CONCLUSION: The rehabilitation timing hypothesis is plausible and clinically testable, but direct evidence in stroke populations is currently lacking. Scheduling motor therapy close to sleep is a low-cost strategy that warrants formal evaluation as a means of enhancing stroke recovery. Formal clinic...

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Oxytocin neurons signal state-dependent transitions from rest to thermogenesis and behavioral arousal

Adam Nelson 2026-10-30 1 min 64 words LOW VISION

Summary: Core body temperature (Tb) is defended within narrow limits through thermoregulatory behaviors like huddling, nesting, and physical activity, as well as autonomic responses like brown fat thermogenesis. While Tb displays regulated fluctuations across different behavioral states and rest/arousal cycl...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Comparison of the efficacy of neuromodulation techniques based on corticospinal excitability in improving exercise-induced fatigue in athletes: a systematic review and network meta-analysis

👤 Li Qing 📅 2026-10-28 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSION: TEAS combined with acupuncture may be associated with the most favorable SUCRA ranking, but this should be interpreted as exploratory for improving EIF in athletes. However, the overall quality of current evidence is low, and SUCRA rankings should be interpreted as exploratory. Future hi...

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Across established non-invasive neuromodulation and emerging deep-targeting strategies: a comparative bibliometric analysis of TI, TMS, tDCS, and tFUS/TUS/LIFU

👤 Shuang Wu
📅 2026-10-28 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Non-invasive brain stimulation (NIBS) is expanding from conventional cortical modulation toward approaches capable of engaging deeper brain targets, but the literature structures of major modalities have rarely been compared within a common framework. We conducted a comparative bibliometric mapping ...

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mPFC-Dependent Valuation Shapes Information Sharing: Causal Evidence from tDCS and a Dyadic Social Media Paradigm

👤 Jinjin Wang 📅 2026-10-28 📖 1 min 📝 72 words TDCS TACS TRNS

Summary: Information sharing drives the spread of both accurate and misleading content, yet its psychological and neural mechanisms remain unclear. We investigated whether perceived self- and social relevance influence sharing through downstream content evaluations, and whether this process depends on medial...

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Joint a priori kinetic modeling with neural ODE for dual-tracer dynamic PET separation

👤 Su Ruan 📅 2026-10-28 📖 1 min 📝 66 words TDCS TACS TRNS

Summary: Simultaneous dual-tracer dynamic PET imaging acquires a time series of images and could streamline oncology workflows by enabling the quantification of complementary biological processes in a single session. However, separating the superimposed tracer signals from the resulting mixed image sequence ...

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A dual diversity perspective on individual response variability in tDCS motor learning research: Some comments on Blake et al., 2026

Thorsten Rudroff 2026-10-29 1 min 2 words TDCS TACS TRNS

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Transcranial Direct Current Stimulation (tDCS) and Smoking Cessation: Clinical and Neurocognitive Outcomes Among Turkish Adults

Wayne M Dinn 2026-10-29 1 min 22 words TDCS TACS TRNS

Summary: CONCLUSION: Unfortunately, in the present modest study, tDCS was not an effective smoking cessation approach among adult smokers in this sociocultural milieu.

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Transcranial alternating current stimulation as psychiatric intervention: A meta-analytic primer

👤 Miles Wischnewski
📅 2026-10-29 📖 1 min 📝 36 words

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Summary: CONCLUSION: This transdiagnostic meta-analysis identifies early therapeutic signals of tACS across psychiatric symptoms, with the most consistent evidence observed for depressive symptoms. Frequency-specific effects may influence clinical outcomes and warrant further investigation to optimize stimul...

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Safety and Efficacy of Anodal transcranial direct-current stimulation over the Dorsolateral Prefrontal Cortex in Post-Stroke Cognitive Impairment: A Systematic Review and Meta-Analysis

👤 Dongyan Wang
📅 2026-10-29 📖 1 min 📝 39 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the DLPFC may improve global cognitive screening performance and ADL in PSCI patients, yet low methodological quality and inadequate safety reporting limit the current evidence. Further high-quality RCTs with rigorous adverse event monitoring are warranted.

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Comparative chloroplast genomics reveal two new species of Primula sect. Auganthus (Primulaceae) from the karst regions of China: Primula shibingensis and P. longmenshanensis

Zhikun Wu
2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSION: Based on morphological and phylogenetic evidence, both newly discovered species are recognized as new to science. They are named Primula longmenshanensis J.L.Gu & Z.K.Wu, sp. nov. and Primula shibingensis J.L.Gu & Z.K.Wu, sp. nov. according to their distribution localities. Their morphol...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu
📅 2026-10-30 📖 1 min 📝 58 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Brain activation changes during working memory training: Individual capacity effects on functional near-infrared spectroscopy measures

👤 Tali Bitan 📅 2026-10-28 📖 1 min 📝 66 words FNIRS

Summary: Working memory (WM) training has shown promise for cognitive enhancement, but the neural mechanisms underlying training effects and offline consolidation processes remain unclear. This study examined the correspondence between behavioral and neural changes during WM training and how individual diffe...

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Dyadic fNIRS Hyperscanning: Experimental Paradigms and What They Reveal About Interbrain Synchrony

👤 Miguel Castelo-Branco 📅 2026-10-28 📖 1 min 📝 65 words FNIRS

Summary: Functional near-infrared spectroscopy (fNIRS) hyperscanning enables the simultaneous measurement of the brain activities of various individuals performing shared or interactive tasks and estimating their interbrain synchrony (IBS). IBS is believed to reflect different mechanisms, including social bo...

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Neural, vascular, motor and autonomic habituation to high-intensity auditory stimulation: A multimodal characterization of the human defensive responses

👤 Carlos M Gómez 📅 2026-10-28 📖 1 min 📝 55 words FNIRS

Summary: The defense reflex has been extensively characterized across sensory modalities; however, the integration of electrical, neurovascular, motor, and peripheral autonomic responses involving the central and autonomic nervous systems remains poorly understood. This study investigated defensive responses...

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Temporal patterns of prefrontal hemodynamics during Tai Chi, Baduanjin, and Liuzijue: a wearable fNIRS study protocol

👤 Qun Fang 📅 2026-10-29 📖 1 min 📝 64 words FNIRS

Summary: Traditional mind-body practices integrate movement, breathing, and attention, but the temporal characteristics of prefrontal hemodynamics during ongoing practice remain unclear. This study protocol aims to compare process-level patterns of prefrontal oxygenation during 24-form Tai Chi, Baduanjin, an...

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Comprehensive artificial intelligence-assisted mental health program for university students in China (CAMPUS): study protocol for a multi-center prospective cohort with planned embedded randomized intervention trials

Yange Wei
2026-10-29 1 min 56 words FNIRS

Summary: University students face a rising prevalence of depression, anxiety, insomnia, and suicidal ideation that exceeds the capacity of conventional, resource-constrained campus support services. Although artificial intelligence (AI)-assisted screening tools have emerged, many have evaluated selected moda...

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From individual responses to dyadic alignment: An fNIRS hyperscanning study of continuous competition

Ali Nikhalat-Jahromi 2026-10-29 1 min 63 words FNIRS

Summary: Competition combines opposing goals with continuous interpersonal dependence, yet its neural basis has been studied mainly using discrete or turn-based paradigms. We examined continuous competition at individual, dyadic, and cross-level scales using a virtual-tennis task. Fifty male university stude...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Susceptibility to task-irrelevant auditory distractors in relation to visual working memory in children with and without attention-deficit/hyperactivity disorder

👤 Yan Song 📅 2026-10-28 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder associated with increased distractibility. Previous research has reported that susceptibility to task-irrelevant visual distraction is linked with visual working memory (WM) capacity in adults. Here, we examin...

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A synchronized dual-scale platform combining flexible micro-ECoG and skull-screw EEG reveals frequency-dependent cortex-to-skull spectral gain in awake mice

👤 Zhaohui Song
📅 2026-10-28 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) is the most scalable window onto brain dynamics, yet how cortical activity is transformed before reaching extracranial electrodes has largely been inferred from forward models rather than directly measured. We built a dual-scale sensing platform in awake mice that record...

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Federated Spiking GANs with Renyi Differential Privacy for privacy-preserving synthetic EEG generation in brain-computer interfaces

Garima Bajwa
2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalography (EEG) data used in brain-computer and brain-machine interfaces (BCI/BMI) contain sensitive, user-specific information, creating privacy risks in consumer applications such as gaming, human augmentation, and virtual reality. Existing methods like encryption and anonymization ei...

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Beyond Static Representations: Learning Electroencephalogram Spatiotemporal Dynamics for Enhanced Emotion Recognition

Wei Wu 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Electroencephalogram (EEG)-based emotion recognition plays a pivotal role in brain-computer interfaces and mental health monitoring. However, conventional approaches often neglect the intrinsic spatial correlations among EEG channels and fail to effectively model both global temporal dependencies an...

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A two-stage RAG-LLM pipeline for roadmapping neuroscience research in operational environments

Cara A Spencer 2026-10-29 1 min 32 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Neuroscientific research and applications in real world environments are often constrained by limited access, equipment limitations, and high costs. Consequently, robustly selecting constructs to focus effort on is a high-impact task.

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Brain-computer interface technology for chronic pain management

Nabi Rustamov 2026-10-29 1 min 54 words BRAIN COMPUTER INTERFACE

Summary: INTRODUCTION: Chronic pain remains inadequately managed by pharmacological and interventional therapies, which carry risks including opioid-related complications and limited long-term efficacy. Brain-computer interfaces (BCI) have emerged as a non-pharmacological therapeutic approach, most commonly ...

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Attention-BrainVerse: A Self-Adaptive Hybrid BCI-VR Interaction Model Driven by Attention

Rui Yang 2026-10-29 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interface (BCI) is an emerging interactive technology that, when integrated with virtual reality (VR), exhibits significant potential for diverse applications. However, widespread adoption of BCI-VR systems remain hindered by challenges such as poor usability, high hardware costs, and...

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Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

Jennifer Collinger 2026-10-29 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally occurring dis...

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Spectral smoothing explains the offline gain from cross-order single-trial MEG movement decoding for non-invasive BCI

Youmin Shin 2026-10-29 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: Acceleration, velocity, and position can be combined in non-invasive movement decoding, but in this dataset only acceleration is measured; velocity and position are deterministic transforms obtained by integration. We therefore tested whether cross-order decoder fusion improves estimation in positio...

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Depth-adaptive suppression of subject-dependent interference for robust cross-subject EEG emotion recognition

Liangxun Shuo 2026-10-30 1 min 16 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Selective depth-aware suppression provides a practical and interpretable strategy for robust EEG-based affective brain-computer interfaces.

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Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

> **Summary:** Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

I'm now maintaining passlib

👤 /u/Amazing_Learn 📅 2026-10-30 📖 1 min 📝 247 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <p>Hi everyone, around a week ago passlib was transferred to me on PyPi after the previous maintainer became unreachable. I've been trying to maintain it in the meantime under a different...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xj74pw/im_now_maintaining_passlib/

Geist Serif

👤 Zm44 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://vercel.com/blog/introducing-geist-serif Comments URL: https://news.ycombinator.com/item?id=50305601 Points: 61 # Comments: 20

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https://vercel.com/blog/introducing-geist-serif

Florilegium: Procedural Flowers in LaTeX

👤 nanna 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/kevinraymond/florilegium Comments URL: https://news.ycombinator.com/item?id=50305637 Points: 74 # Comments: 11

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https://github.com/kevinraymond/florilegium

AI isn't replacing SREs, it's deskilling them

👤 dorianniemiec 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

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US tech stocks see sharp falls over AI cost fears

👤 mindracer 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.bbc.co.uk/news/articles/cvwy72e0n72yo Comments URL: https://news.ycombinator.com/item?id=50306008 Points: 181 # Comments: 160</...

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A Working Nintendo in ~1GB of Qwen Weights

👤 dbreunig 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/dbreunig/nesformer Comments URL: https://news.ycombinator.com/item?id=50306442 Points: 87 # Comments: 79

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https://github.com/dbreunig/nesformer

Show HN: Spinf-serve – inference server for bulk scoring and decisions

👤 giarec 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/spinfinc/spinf-serve Comments URL: https://news.ycombinator.com/item?id=50306815 Points: 7 # Comments: 0

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https://github.com/spinfinc/spinf-serve

Proton's CEO Cancels AI 'Re-Interviews'

👤 cdrnsf 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.404media.co/protons-ceo-cancels-ai-re-interviews/ Comments URL: https://news.ycombinator.com/item?id=50307183 Points: 135<...

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https://www.404media.co/protons-ceo-cancels-ai-re-interviews/

My favourite PowerMac is a Nintendo Wii

👤 DHowett 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://virtuallyfun.com/2026/10/29/my-favourite-powermac-is-a-nintendo-wii/ Comments URL: <a href="https://news.ycombinator.com/item?id=50307190">https://news.ycombinator.com/item?id=50...

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https://virtuallyfun.com/2026/10/29/my-favourite-powermac-is-a-nintendo-wii/

Anthropic may not secure investment-grade rating after IPO, Blackstone says

👤 petethomas 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.ft.com/content/59383c7f-8d82-493f-bf01-d2e862a1018f Comments URL: https://news.ycombinator.com/item?id=50307219 Points: ...

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GitHub Incident with Actions, Pages, and Webhooks

👤 damontal 📅 2026-10-30 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.githubstatus.com/incidents/pwxsjgnvrk6b Comments URL: https://news.ycombinator.com/item?id=50307723 Points: 39 # Comments: 19...

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https://www.githubstatus.com/incidents/pwxsjgnvrk6b

Automatic computational reproducibility checks with Metacheck

noreply@blogger.com (Daniel Lakens)
2026-10-30 19 min 3854 words

TWENTY PERCENT STATISTICIAN

Summary: When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

Steve Luck
2024-03-16 1 min 106 words

ERP BOOT CAMP

Summary: The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

> **Summary:** The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

📖 1 min 📝 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Neurodegenerative diseases in Sub-Saharan Africa: diagnostic invisibility, biological heterogeneity and a precision-neuroscience agenda

📖 1 min 📝 11 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maulilio John Kipanyula

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006962?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

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https://www.sciencedirect.com/science/article/pii/S0306452226006949?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Algorithmic Geometry Shapes the Biological Meaning of Thalamocortical Structural Covariance

👤 Liu, Y., Sydnor, V. J., Dai, A., Ziolkowski, J., Fu, S., Devenyi, G. A., Satterthwaite, T., Chakravarty, M., Yee, Y.

📅 2026-10-30 📖 1 min 📝 306 words BIORXIV NEUROSCIENCE

Summary: Neuroimaging studies often derive and discuss structural covariance (SC) as a representation of underlying brain circuitry that reflects coordinated maturation, anatomical connectivity, transcriptional similarity, and shared genetic influences. These associations motivate its use to study brain orga...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762356v1?rss=1

Item-specific clustering of high gamma-band activity on theta phase during speech categorization

👤 Ten Oever, S., Reithler, J., Archila-Melendez, M. E., Gommer, E., Rouhl, R., Kranen-Mastenbroek, V., Roberts, M. J., Peters, J.

📅 2026-10-30 📖 1 min 📝 143 words BIORXIV NEUROSCIENCE

Summary: Theta-gamma phase-amplitude coupling (PAC) has been proposed as a mechanism for separating neural representations in time. Although item-specific PAC has been reported, its role in perception remains unclear. Notably, previous work has shown that theta phase can bias perception, raising the question...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.29.762424v1?rss=1

Disrupting SORL1-retromer endosomal recycling in mice drives the age-dependent neurodegenerative signature of Alzheimer's disease

Qureshi, Y. H., Guo, J., Rinald, J. H., Li, Z., Soni, R. K., Govindarajan, A., Tortora, E., Uzunova, G., Yang, Y., Kannan, S., Nanayakkara, P., Troy, C. M., Small, S. A.

2026-10-30 1 min 157 words BIORXIV NEUROSCIENCE

Summary: MRI studies have mapped the atrophic signature that distinguishes Alzheimer's disease (AD), isolating regions differentially vulnerable to its age-dependent neurodegenerative process. We test the hypothesis that an endosomal recycling pathway regulated by the interaction of SORL1 with retromer, whic...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.26.761980v1?rss=1

Generation of human iPSC-derived motor neurons with defined regional identities across the rostrocaudal axis

Ning, K., Tindel, I., Singh, B. K., Zogu, B., Ho, R.

2026-10-30 1 min 205 words BIORXIV NEUROSCIENCE

Summary: This study establishes a unified pipeline for generating human induced pluripotent stem cell (iPSC)-derived motor neurons (iMNs) with defined hindbrain, cervical, and lumbar identities to support anatomically resolved studies of motor neuron development and degeneration. Conventional differentiation...

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https://www.biorxiv.org/content/10.64898/2026.10.27.761477v1?rss=1

Brain representations of visual short-term memories during healthy human aging: occipital decline and frontoparietal compensation

Mitchell, D. J., Cam-CAN,, Cusack, R.
2026-10-30 1 min 150 words

BIORXIV NEUROSCIENCE

Summary: Visual short-term memory declines steeply with age, but the link to neural representations of memory content remains unknown. We used fMRI decoding to relate remembered motion-direction representations in visual and frontoparietal cortex to age and performance, in 115 adults, aged 25-87, from the Ca...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761884v1?rss=1

Lexical surprisal drives pupil dilation during continuous story listening

Terzo, C., Berlot, E., de Lange, F.
2026-10-30 1 min 236 words

BIORXIV NEUROSCIENCE

Summary: The pupil famously adjusts its dilation in response to light, but its role extends far beyond this basic sensory function. Its fluctuations index arousal and mental effort, beyond luminance-related dynamics, making it a sensitive marker of internal cognitive states. A growing body of work shows that...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.26.761911v1?rss=1

Salience network expansion as a transdiagnostic neural marker for psychopathology risk in adolescence

👤 Koirala, S., Hermosillo, R., Godfrey, K. J., Anderson, M., Lundquist, J., Moser, J., Feczko, E., Miranda Dominguez, O., Weldon, K. B., Linder, I., Mahmoudi, M., Uddin, L. Q., Nelson, S. M., Elison, J. T., Fair, D., Tervo-Clemmens, B.

📅 2026-10-30 📖 1 min 📝 216 words BIORXIV NEUROSCIENCE

Summary: Increased cortical surface area of the brain's salience network (SAL) has been observed in depression and appears to emerge during early adolescence. However, similar expansion has also been reported in bipolar disorder (BD), obsessive compulsive disorder (OCD), and other atypicalities of salience n...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762645v1?rss=1

Glutamatergic and Redox Abnormalities in Major Depressive Disorder: Links to Childhood Maltreatment and Perceived Stress

👤 Syed, S., Dan, R., Hutson, L. M., Brown, A. R., Wang, C., Stein, A., Chen, X., Du, F., Pizzagalli, D. A.

📅 2026-10-30 📖 1 min 📝 244 words BIORXIV NEUROSCIENCE

Summary: Background: Childhood sexual abuse (CSA) increases the risk for major depressive disorder (MDD), yet the underlying neurobiological mechanisms remain poorly understood. We examined (1) associations of medial prefrontal cortex (mPFC) glutamatergic metabolites and redox balance with childhood maltreat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762615v1?rss=1

Measuring conscious tactile perception without report: pupil, blink, and microsaccade signatures of a tactile no-report paradigm

Sharma, A. A., Senjalia, R., Woodworth, A., Jin, D., Aerts, S., Cormier, D., Khalaf, A., Blumenfeld, H.

2026-10-30 1 min 219 words BIORXIV NEUROSCIENCE

Summary: Conscious perception is often inferred from behavioral reports, which conflate awareness-related neural signals with the motor and decision processes required to respond. No-report paradigms address this by using spontaneous physiological signals, such as eye movements, to infer perception. Although...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.29.762884v1?rss=1

Theta-Phase Targeted Encoding Gates Memory Sensitivity

Raissle, A., van de Ven, V., Leunissen, I., Fang, Z., Valente, G., Naseri, P., Ten Oever, S.

2026-10-30 1 min 170 words BIORXIV NEUROSCIENCE

Summary: Theta phase has been proposed to play an important role in memory. It may gate encoding and retrieval, such that certain phases are more favorable for memory processing than others, or become part of the memory representation, such that reinstating the encoding phase benefits retrieval. To test and ...

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https://www.biorxiv.org/content/10.64898/2026.10.29.762605v1?rss=1

Spatial single-cell mapping of transcriptional differences across genetic backgrounds in mouse brains

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41592-026-03289-2

Assessing whole-cortex excitability from electromagnetic brain signals

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11005-1

Establishing a normative model for turbulence dynamics in the brain

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77914-7

Large-scale EEG reveals shared and distinct neural oscillatory features across major psychiatric disorders

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41398-026-04532-2

Transformer language models reveal distinct patterns in aphasia subtypes and longitudinal change

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77518-1

Publisher Correction: Regional heterogeneity in phenotypic and genetic associations between bone and brain in humans

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79235-9

A task-fMRI study in adolescent and adult survivors of childhood cancer

📅 2026-10-30 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Head stabilization patterns reflect long-term postural stability alterations during gait associated with history of concussion

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https://www.nature.com/articles/s41598-026-78672-2

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

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📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

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https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

Bocheng Wang 2026-10-07 1 min 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

Elif Hocaoglu 2026-10-15 1 min 215 words FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

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Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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Cerebral blood flow asymmetry in motor-related regions is associated with 30-day upper-limb motor recovery after acute ischemic stroke

Xin Wang 2026-10-30 1 min 337 words FRONTIERS NEUROSCIENCE

Summary: BackgroundUpper-limb motor impairment is a leading cause of long-term disability after acute ischemic stroke (AIS), yet early prognostication relies mainly on clinical scales. Cerebral blood flow (CBF) asymmetry measured by arterial spin labeling (ASL) perfusion MRI may indicate recovery potential. ...

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Cross-subject EEG driving fatigue detection: early personal spectral reference and causal long-sequence residual encoding

Lanlan Lv 2026-10-30 1 min 265 words FRONTIERS NEUROSCIENCE

Summary: BackgroundIn cross-subject electroencephalography (EEG)-based driving-fatigue detection, most existing decoders are designed around second-scale short windows, whereas fatigue is a slow drift; short-window spectral features provide little direct evidence of that slow drift, limiting fatigue decision...

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Glial handling of alpha-synuclein seeds: uptake, degradation and redistribution in pathological propagation

R. M. Meade 2026-10-30 1 min 202 words FRONTIERS NEUROSCIENCE

Summary: α-synuclein (αS) aggregation and propagation are central features of Parkinson's disease pathology. Preformed fibril (PFF) models have shown that extracellular αS aggregates can enter neurons and seed aggregation of endogenous αS. Within the brain, these seeds also encounter microglia and astrocytes...

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A progenitor with immature Bergmann glia-like characteristics in the developing cerebellar cortex

Kazuhiko Sawada 2026-10-30 1 min 203 words FRONTIERS NEUROSCIENCE

Summary: The cerebellum is an ancient brain component, conserved across mammalian species, that has classically played roles in regulating muscle tone, limb coordination, and equilibrium, as well as in non-motor functions, such as language, cognition, and emotion. Recently, we reported Nestin-expressing cell...

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Case Report: A novel hemizygous nonsense variant in PPP1R3F gene associated with developmental delay, intellectual disability and autism spectrum disorder

Jingsi Luo 2026-10-30 1 min 328 words FRONTIERS NEUROSCIENCE

Summary: BackgroundPPP1R3F deficiency syndrome is an extremely rare, recently identified X-linked recessive neurodevelopmental disorder. It is caused by hemizygous variants in the PPP1R3F gene, which encodes a glycogen-targeting subunit of the brain-specific protein phosphatase 1 (PP1). Dysfunction of PPP1R3...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1998834

Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

Seho Lee 2026-10-21 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

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